PATIENT RECORD SYSTEM WITH DISEASE TRACKING AND ARIMA-BASED FORECASTING FOR TOP 5 DISEASES IN LOS BAÑOS, LAGUNA

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In partial fulfilment of the requirement for the degree
Bachelor of Science in Information Technology
Specialized in *Web and Mobile Applications Development*

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APPROVAL SHEET

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The Researchers

DEDICATION

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R.E.B.F

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ABSTRACT

This study developed a Patient Record System with Disease Tracking and ARIMA-Based Forecasting to address the challenges faced by Rural Health Units (RHUs), where manual and paper-based processes often delay reporting and hinder effective disease monitoring. The system provides a web-based platform that enables health workers to manage patient records, visualize disease cases through GIS mapping, and generate short-term forecasts using ARIMA to help anticipate possible increases in measles, chickenpox, hepatitis, tuberculosis, and dengue. Built through the Agile Scrum methodology, the system integrates modules for patient management, tracking, reporting, forecasting, and secure role-based access, and was evaluated using ISO 25010 by IT experts and the Technology Acceptance Model (TAM) by RHU staff, both yielding strong ratings in functionality, usability, reliability, and overall acceptance. These results indicate that the system is suitable for real RHU environments, supporting organized record-keeping, improved disease surveillance, and timely planning, while also aligning with the Sustainable Development Goals, particularly those promoting good health, innovation, and sustainable communities.

Keywords: *Patient Record System, Disease Tracking, ARIMA Forecasting, Diseases, Health Information System, GIS Mapping, Rural Health Unit (RHU).*

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DEFINITION OF TERMS

Admin	Refers to the authorized user who manages the overall operation of the system. The admin is responsible for creating staff accounts, monitoring system activities, and maintaining system configuration and access control.
ARIMA (Autoregressive Integrated Moving Average).	A statistical forecasting model used to analyze and predict trends based on historical data. In this study, ARIMA is applied to forecast the possible spread of diseases using recorded patient data.
Diseases	Health conditions recorded and monitored in the system, including measles, chickenpox, hepatitis, tuberculosis, and dengue, based on reported cases in the Rural Health Unit.
Forecasting	The process of predicting future trends or occurrences using past and present data. In this study, forecasting is used to estimate potential disease outbreaks through the ARIMA model.
Geographic Information System (GIS)	A digital tool that collects, manages, and analyzes spatial and geographic data. In the project, GIS is integrated to map and visualize disease cases across specific locations.
Health Information System (HIS)	A system designed to collect, store, manage, and transmit health-related information within a healthcare organization. It supports data-driven decisions and

	efficient patient record handling.
Patient Record System	The core system developed in this study that allows health workers to manage patient data digitally, including patient registration, medical history, diagnosis, and treatment details.
Rural Health Unit (RHU)	A local government health facility responsible for providing primary healthcare services to communities, particularly in rural areas. The RHU serves as the main context of the study for system deployment and evaluation.

CHAPTER I

INTRODUCTION

Project Context

Diseases remain among the world's major public health issues, particularly in developing nations such as the Philippines. Infectious diseases are a leading cause of morbidity and mortality in Southeast Asia, as stated by the World Health Organization (2024), and outbreaks are likely to overwhelm the health facilities of local health systems. Infection transmission of diseases continues to pose a public health concern by complicating surveillance, reporting, and case control, according to the Department of Health (2023). The lack of a real-time surveillance system and poor record-keeping in the majority of rural settings limit early intervention and effective outbreak control.

In the Philippines, Rural Health Units (RHUs) and health centers often use manual, paper-based systems to record patient information and track disease cases. A study by Acacio-Claro et al. (2023) found that over 60% of RHUs in the Philippines use manual systems, which are susceptible to data loss, delayed reporting, and errors in disease monitoring. This inefficiency causes delayed response times and difficulties in controlling disease outbreaks, particularly in highly populated or remote areas, according to the study by Ongkeko et al. (2016). With the country’s archipelagic geography and resource limitations, there is an urgent need for digital health innovations to support healthcare workers and improve data management.

New technological developments have shown potential in bridging these gaps. The Philippine Council for Health Research and Development (2022) has identified that the application of Geographic Information System (GIS) tools

enhanced disease distribution tracking and case mapping among regions. GIS allows for spatial data visualization by healthcare workers, which helps in faster decision-making amidst outbreaks. According to López et al. (2021), ARIMA-based models can predict infectious disease trends through time-series forecasting that analyses historical data patterns and identifies potential future increases or decreases in case numbers.

Building on these developments, this study proposes a Patient Record System with Disease Tracking and ARIMA-Based Forecasting for Top 5 Diseases in Los Baños, Laguna. The project aims to digitize patient records, monitor priority disease cases, and predict future trends of the top five reported diseases, namely measles, chickenpox, hepatitis, tuberculosis, and dengue, using an ARIMA-based forecasting model. As emphasized by Chadsuthi et al. (2018), having an organized and reliable system for recording and monitoring health data can improve the response time and management of disease outbreaks. The project, currently at the concept and paper proposal stage, is intended for future implementation in Rural Health Units (RHUs) to help address problems such as manual reporting, data delays, and incomplete patient records.

As demonstrated by similar digital health initiatives in Southeast Asia, adopting health information systems not only streamlines data management but also strengthens community-level disease surveillance based on the study of Lo et al. (2016). This study contributes to the expanding field of health informatics research in the Philippines by offering a specialized platform tailored to local healthcare needs. By combining disease tracking, GIS mapping, and ARIMA forecasting, the system aims to provide a reliable, accessible, and

efficient tool for improving public health response to commonly occurring diseases in local communities.

Project Objectives

The primary objective of this study is to develop a Patient Record System with Patient Tracking and ARIMA-Based Forecasting for Top 5 Diseases recorded in the Rural Health Unit, that aims to improve health record management, patient monitoring, and disease trend forecasting in rural health units and local health centers.

Specifically, it aims:

1. To develop a secure patient record system that allows convenient recording, storing, and retrieving of patient information while keeping the data protected from unauthorized access.
2. To implement a map-based disease tracking that helps healthcare staff view the locations of reported cases using available patient data.
3. To develop an ARIMA-based forecasting model that utilizes recorded case data to predict short-term future case trends for the top five recorded diseases.
4. To evaluate the developed system using the ISO 25010 quality characteristics and to assess user acceptance based on the Technology Acceptance Model (TAM).

Project Purpose

This project aims to develop a Patient Record System with Disease Tracking and ARIMA-Based Forecasting for Top 5 Diseases, designed to modernize patient record keeping and disease monitoring in rural health units and local health centers. The system supports several Sustainable

Development Goals (SDGs) by implementing a web-based health information management system with map-based tracking and forecasting features. The system focuses on Goal 3: Good Health and Well-Being, by improving healthcare services through efficient surveillance and timely patient data management; Goal 9: Industry, Innovation, and Infrastructure, through digital health technology development; and Goal 11: Sustainable Cities and Communities, through technology-assisted public health management. By focusing on the top five prevalent diseases in the community, the system contributes to building safer, more resilient, and better-prepared local health environments.

Rural Health Units and Local Health Centers. The system offers a map-based disease tracking feature that allows these facilities to visualize the distribution of disease cases within their locality. Through this, they can monitor disease trends in a specific area based on recorded data, helping them allocate resources and implement localized health interventions.

Health Authorities and Decision-Makers. With the inclusion of an ARIMA-based disease forecasting model, the system can support decision-makers by providing data-driven insights and forecasts of possible case trends based on previously recorded data. This can aid in developing preventive measures and health programs aligned to the needs of their community.

Future Researchers. This study serves as a foundational reference for future researchers interested in developing medical information systems and disease tracking applications. It can also provide ideas for enhancing patient data security, health surveillance, and the use of ARIMA-based models in public health informatics.

Scope and Limitation of the Study

This study focuses on the development of a Patient Record System with Disease Tracking and ARIMA-Based Forecasting for Top 5 Diseases. The system is intended for use in rural health units and local health centers to improve the recording, storage, and management of patient health records. It is designed to accommodate and store five (5) years' worth of collected data from the Rural Health Unit (RHU) as reference for monitoring and record management. The system enables healthcare staff to maintain patient information securely, while providing a map-based tracking feature that visualizes the locations of top disease cases based on patient data. The system focuses five common diseases frequently recorded in local health centers in Los Baños, which include measles, chickenpox, hepatitis, tuberculosis, and dengue. The web system has different modules and features.

Account Module

This module controls user access to the system. It provides healthcare staff with the ability to log in, update profile data, and safely log out. The system includes role-based access controls that permit access to only information relevant to the role assigned in the healthcare organization. User authentication controls provide confidentiality and integrity of sensitive health information

Patient Module

Patient Module enables the healthcare staff to create detailed patient records, view patient records, insert new patients into the system, update patient records, and view consultation history. The module stores critical patient information including personal, demographic, medical history, diagnosis details, and treatment plans.

Disease Tracking Module

This module facilitates the recording, monitoring and analysis of recorded disease cases. Healthcare staff can record new patient cases, update the status of existing cases, and conduct patient tracking activities. The map view feature provides geospatial visualization of disease spread, enabling health officials to identify clusters and potential hotspots. This visual representation helps in monitoring disease patterns and informing targeted intervention strategies.

Reporting Module

The Reporting Module generates analytics and reports on disease prevalence, patient demographics, and treatment outcomes. Users can generate various types of reports, export data for further analysis through dashboards, and receive automated alerts about critical events such as outbreak levels are reached or significant changes in disease patterns.

ARIMA-Based Analytics

This advanced module utilizes an ARIMA-based forecasting model to analyze collected data and provide predictive insights. It incorporates time-series forecasting capabilities to predict potential changes or trends in disease cases based on historical and current data. Geographic Information System (GIS) analysis integrates spatial data with health records to identify geographical patterns of disease distribution. The risk assessment component evaluates population vulnerability to specific diseases, while continuous updates to the ARIMA model improve forecast accuracy as new data becomes available.

CHAPTER II

THEORETICAL FRAMEWORK OF THE PROJECT STUDY

Review Related Literature, Studies, and System

The literature review provides a broad understanding of the existing technologies and innovations relevant to the development of health informatics systems, particularly those involving patient record management, disease case tracking, and forecasting. This section discusses various studies and systems related to Electronic Health Records (EHR), Disease Tracking Systems, GIS-based Disease Mapping, ARIMA-based Forecasting in Disease Trend Analysis, and Health Information Systems in Rural Health Units (RHU).

Electronic Health Record

According to Aliabadi et al. (2020), electronic health records (EHRs) are computerized versions of patient medical records that capture, store, and create health information electronically. In their systematic review on EHR-based disease surveillance systems, they emphasized that EHRs serve as essential tools for integrating clinical documentation and supporting public health monitoring. EHR systems are made to automate clinical documentation, enhance the quality of care, facilitate tracking of healthcare utilization, and ease billing and coding. Though their benefits are evident, use of EHR systems by healthcare providers has been slow, partly due to technical and interoperability challenges. Aliabadi et al. (2020) also noted that the successful implementation of EHRs depends on the proper integration of software, hardware, and IT networks that ensure secure and efficient information exchange between healthcare facilities. Through enhanced patient data management, enhanced clinical processes, and enhanced delivery of healthcare, electronic health

records, or EHRs, have transformed healthcare today. A study by Campanella et al. (2016) asserts that by increasing compliance with clinical guidelines, reducing medical mistakes, and enhancing productivity, utilization of EHRs has significantly increased the quality of healthcare. Equally, EHR systems assist in enhancing patient safety by minimizing medication-related adverse events and facilitating communication among providers, as Alotaibi and Federico (2017) observe. General success of EHR implementation is, nonetheless, marred by chronic issues of interoperability challenges, breaches, and physician burnout, according to Kruse et al. (2018). Further, by granting patients access to their EHRs, Neves et al. (2020) described how patient empowerment to take part in their own care and decision-making enhances the quality of care.

Disease Tracking Systems

Disease tracking systems play a crucial role in increasing the efficiency of healthcare delivery through real-time monitoring and enhanced documentation. According to Zhao et al. (2023), modern disease tracking systems leverage artificial intelligence, cloud computing, and Internet of Things (IoT) technologies to improve surveillance and enable faster detection of disease outbreaks. These systems have significantly enhanced the accessibility and timeliness of health data, reducing delays in healthcare response and decision-making. Butler et al. (2020) observe that the integration of electronic health records (EHR) and disease tracking enables smooth integration of data, hence reducing redundancy and maximizing patient-centered care. Shafi et al. (2024) argue that cloud-based and forecasting-based tracking systems have transformed disease monitoring, especially during

health crises such as the COVID-19 pandemic. Mahmmod et al. (2024) further affirm this argument, observing that modern tracking systems using predictive models such as the Autoregressive Integrated Moving Average (ARIMA) can provide real-time trend analysis and short-term disease forecasting that supports proactive health response. Said et al. (2024) suggest the use of blockchain technology as an effective remedy to enhancing the security and accessibility of disease records, hence eliminating issues of unauthorized access and data leakage. Bhatia (2023) investigates how digital tracking innovations help reduce the issue of hospital overcrowding and enhance overall patient outcomes. These studies affirm the significance of disease tracking systems in contemporary healthcare, with emphasis on their critical role in monitoring diseases, predicting possible outbreaks, and formulating timely response strategies.

GIS-Based Disease Mapping

Geographic Information System (GIS)-based methods for disease mapping and monitoring of patients have emerged as fundamental tools in contemporary public health management, facilitating disease surveillance and management of outbreaks. Brown et al. (2022) emphasize that GIS-based monitoring enhances decision-making efficacy during outbreak management through the supply of real-time geospatial data, thereby increasing the capability of health authorities to detect, analyze, and control the spread of diseases efficiently. According to Lee, Park, and Kim (2023), GIS facilitates early disease detection through the combination of epidemiological data with demographic and environmental variables, leading to more insightful decision-making at local, regional, and national levels. GIS technology facilitates

effective multi-disease surveillance through wide coverage of diverse diseases, enabling healthcare specialists to visualize instances of diseases, forecast transmission modes, and optimize resource allocation. The same functionality assists in patient monitoring to facilitate timely and focused provision of health interventions, thereby reducing the severity of disease outbreak impact. The integration of GIS with electronic patient tracking systems can optimize emergency response through tracking of patient location and contacts, potentially reducing exposure hazards to disease in the healthcare environment. Photis (2016) asserts that health geography gives a spatial map of the incidence of disease with its corresponding connection to environmental factors. Policymakers are able to control healthcare demand, supply, and healthcare service availability in areas using GIS, enhancing access and resource allocation. In addition, GIS-based disease mapping has been increasingly used to scrutinize public health systems and advance the allocation of resources. Photis (2016) carried out an evaluation of the Greek Public Health Care System between the years 1997–2007, utilizing GIS to map the demand and supply of care by computing hospital, bed, and patient distribution location quotients. The study revealed that GIS mapping allows spatial representations to be created that facilitate complex data, providing valuable knowledge for healthcare policy and planning. Ahmad, Akhtar, and Li (2024) describe how current GIS tools allow researchers to automatically calculate proximity between potential environmental risk sites and disease cases, facilitating spatial searches where distance serves as a measure of exposure. Their findings suggest that these modern GIS platforms are effective

tools for spatial analysis, offering significant potential for disease monitoring and environmental health research.

ARIMA-Based Forecasting for Disease Trend Analysis

Forecasting models play an important role in predicting the spread and behaviour of diseases. The Autoregressive Integrated Moving Average (ARIMA) model is one of the most commonly used approaches for analyzing time-series data to predict disease trends based on historical case records. Galvez and Tarepe (2023), in their study Predicting Dengue Outbreaks in Cagayan de Oro, Philippines Using Facebook Prophet and the ARIMA Model, applied the ARIMA model to forecast weekly dengue cases from 2016 to 2022. The results showed that the ARIMA model provided accurate and reliable predictions that can help local health units prepare for possible outbreaks. Their findings support the feasibility of ARIMA forecasting in the Philippine setting, particularly for dengue and other vector-borne diseases.

Maulion, Destura, and Bantang (2016) used a Seasonal ARIMA (SARIMA) model to forecast dengue cases in Western Visayas from 2008 to 2013. The study showed that SARIMA effectively captured seasonal patterns and could predict peaks in disease occurrence. The researchers explained that this type of forecasting model can be used for preventive health planning and resource allocation during disease outbreaks. This supports the use of ARIMA-based forecasting in improving early warning systems and preparedness in local health facilities.

Naher et al. (2022), in their study Forecasting the Incidence of Dengue in Bangladesh – Application of Time Series Model, compared several forecasting models such as ARIMA, ETS, and TBATS to determine which

model performs best in predicting monthly dengue cases. The ARIMA (2,1,2) model showed the most accurate results, proving its reliability for analyzing disease trends. The study also emphasized the importance of regularly evaluating and updating forecasting models to maintain accuracy and consistency in predicting disease movements.

Fuller, Ssekitoleko, and Kyozira (2025) conducted a study comparing ARIMA and Facebook Prophet models in predicting malaria incidence in Uganda. The ARIMA model achieved higher accuracy in short-term forecasts, while the Prophet model worked better for long-term projections. Their findings confirmed that ARIMA remains one of the most effective models for real-time disease forecasting and can assist public health authorities in timely decision-making and outbreak response.

Turija et al. (2024) developed a forecasting approach that combined the SARIMAX model, which is an extension of ARIMA, with Geographic Information System (GIS) data to predict dengue cases in the Philippines. The integration of time-series forecasting with geospatial mapping improved the accuracy of identifying affected areas and visualizing the spread of diseases. This is closely related to the system developed in this study, which also uses ARIMA-based forecasting together with GIS for monitoring and predicting disease cases.

These studies show that ARIMA-based forecasting is an effective method for predicting the trends of recorded diseases. Its application in various countries and local health studies demonstrates its reliability and adaptability for disease prediction. In this project, the combination of ARIMA forecasting and GIS visualization provides a data-driven approach for monitoring disease

spread, improving health response, and supporting decision-making in Rural Health Units.

Health Information Systems in Rural Health Units (RHUs)

Health Information Systems (HIS) are critical assets in the provision and administration of healthcare services, especially in rural settings where there is limited access to resources. According to Aceron and Garde (2020), states that the situation with the implementation of HIS in the Philippines is hampered by old infrastructure, dispersed data systems, and a lack of trained professionals. These are more pronounced in rural health units (RHUs), where paper-based record-keeping is predominant, hampering the effectiveness of health service delivery. The authors highlight that without major breakthroughs in digital infrastructure and capacity development, RHUs will continue to have difficulties with right data collection and health tracking.

A study by Santos et al. (2024), explored the health system responsiveness of RHUs in the Cagayan Valley Region and found that while health workers are aware of the importance of digital tools, many facilities still lack fully functioning HIS. The study revealed that weak information systems often result in delays in data reporting and difficulty in tracking patient histories. According to the authors, improving HIS infrastructure would not only streamline administrative processes but also enhance the quality of care, especially in marginalized and hard-to-reach communities. To address these challenges, several technology-driven initiatives have been introduced to support RHUs. According to Quilon and Singun (2019), a teleconsultation system developed for RHUs in Metro Vigan provided an innovative way to connect rural patients with health professionals, reducing the need for travel

and improving access to medical advice. The study noted that integrating such systems into existing RHU workflows not only improved access to care but also enhanced the management of patient records and consultations. These results indicate that telehealth can be an important part of a more integrated HIS strategy in rural areas.

Another important innovation in rural HIS is the deployment of the Wireless Access for Health (WAH) program. A study by Arnaout et al. (2014), examined the impact of the WAH electronic health record system on RHUs in the Philippines and found that it greatly improved the efficiency and accuracy of health data management. According to the authors, digitalizing patient records allowed health workers to retrieve data quickly, generate reports with ease, and provide more consistent follow-ups for patients. The success of this program highlights the potential of scalable, technology-based solutions in improving rural health services.

Despite these promising efforts, sustaining and expanding HIS in rural areas remains a challenge. According to Aceron and Garde (2020), many HIS projects struggle to maintain momentum due to limited funding, lack of policy continuity, and low stakeholder engagement. The authors argue that long-term success depends on strong government leadership, inter-agency coordination, and inclusive planning that considers the specific needs of rural communities. Strengthening HIS in RHUs is a crucial step toward equitable health service delivery and improved public health outcomes in underserved regions.

Synthesis and of the Study

The foreign and local literature provided essential insights that support the development of the system. Various studies highlighted the ongoing challenges in disease surveillance and patient data management in rural settings, particularly due to manual processes and delayed reporting (Acacio-Claro et al., 2023; Ongkeko et al., 2016). The integration of Geographic Information Systems (GIS) in healthcare has shown potential in improving outbreak tracking and forecasting (Carvajal et al., 2018; Chadsuthi et al., 2018). Recent studies also demonstrated the use of ARIMA-based forecasting models in predicting disease trends accurately using historical data, supporting data-driven decision-making in public health management (Galvez & Tarepe, 2023; Naher et al., 2022). These findings emphasize the need for a system that digitizes patient records, supports case monitoring, and assists in predicting disease trends through ARIMA-based forecasting, making the project relevant and timely for enhancing public health response and planning.

Table 1. Literature Matrix

AUTHORS	YEAR	TITLE	FINDINGS
Maulion, M. A., Destura, M. D., & Bantang, J.	2016	Forecasting Dengue Cases in Western Visayas Using a Seasonal ARIMA (SARIMA) Model	The study revealed that SARIMA effectively captured seasonal patterns and accurately predicted dengue peaks, supporting its use for early warning and preventive health planning.
Quilon, E. R. C., & Singun, A. P.	2019	Teleconsultation system for rural health units (RHUs) for Metro Vigan, Philippines	It was reported that teleconsultation improves access to care and enhances documentation in rural areas.
Aliabadi, M., Pahl, C., & Hederman, L.	2020	Electronic Health Record–Based Disease Surveillance Systems: A Systematic Review on Challenges and Solutions	The study emphasized that EHR systems play a crucial role in disease surveillance and clinical documentation integration, but face interoperability and implementation challenges.
Butler, J. M., Gibson, B., Lewis, L., Reiber., et al.	2020	Patient-centered care and the electronic health record: Exploring functionality and gaps	The study highlighted that combining tracking systems with EHRs supports continuity and coordination of care.

Neves, A. L., Freise, L., Laranjo, L., Carter, A. W., Darzi, A., & Mayer, E.	2020	Impact of providing patients access to electronic health records on quality and safety of care: a systematic review and meta-analysis	It was observed that allowing patients to access their EHRs promotes engagement and improves health outcomes.
Brown, W., Smith, J., & Harris, M.	2022	GIS-Based Disease Tracking Enhances Decision-Making in Outbreak Response	It was found that GIS enables rapid spatial analysis and enhances disease surveillance.
Naher, N., et al.	2022	Forecasting the Incidence of Dengue in Bangladesh – Application of Time Series Model	Results showed that ARIMA (2,1,2) achieved the best accuracy in predicting monthly dengue incidence, emphasizing ARIMA's reliability for disease forecasting.
Zhao, Y., Zhang, L., & Chen, X.	2023	Artificial Intelligence–Enabled Disease Tracking and Surveillance: A Systematic Review of Technologies and Applications	The research highlighted that AI-based disease tracking enhances detection accuracy and supports real-time outbreak forecasting.
Lee, J., Park, K., & Kim, S.	2023	Advances in GIS-Based Infectious Disease Surveillance: Integrating Spatial Analytics and Epidemiological Modeling	This study showed that integrating GIS with epidemiological data improves early detection and localized response to outbreaks.
Bhatia, A. K.	2023	Digital Tracking Innovations and Their Role in Improving Hospital Management Efficiency	The study found that digital tracking tools reduce hospital congestion and enhance patient flow efficiency.
Galvez, R. C., & Tarepe, R. M.	2023	Predicting Dengue Outbreaks in Cagayan de Oro, Philippines Using Facebook Prophet and the ARIMA Model	The study demonstrated that the ARIMA model effectively forecasted dengue cases based on historical data, proving its reliability for outbreak prediction in local contexts.
Ahmad, M., Akhtar, A., & Li, J.	2024	GIS-Based Applications in Infectious Disease Control: A Review of Current Approaches and Emerging Technologies	The study found that modern GIS tools enhance disease surveillance through spatial correlation analysis, aiding in exposure tracking and outbreak management.
Mahmmod, A. M., Al-Khatib, R., & Hameed, F.	2024	Predictive Tracking and Forecasting Systems in Digital Healthcare	The study introduced an intelligent system combining ARIMA forecasting with IoT-enabled data collection, improving prediction accuracy and health resource planning.
Santos, J. P., Abad, J. C., Aldea, A. A., Catacutan, K. J. A., Itay, S. D., Laureta, V. V. O., & Turingan, R. A.	2024	Health system responsiveness of rural health units in the Cagayan Valley Region: A cross-sectional study	The research observed that inefficiencies in rural HIS delay health responses and patient monitoring.
Turija, J., Gonzales, K., Espinosa, R., Presilda, M., Arcilla, R., & Santiañez, L.	2024	Forecasting Analysis of Dengue Cases with Geospatial Mapping Using SARIMAX Algorithm	The study combined ARIMA-based forecasting with GIS mapping to enhance accuracy in identifying spatial disease trends, supporting integrated data analysis for outbreak control.

Fuller, R., Ssekitoleko, P., & Kyozira, J.	2025	Comparison of ARIMA and FB-Prophet Time Series Models for the Prediction of National and Regional Malaria Incidence in Uganda	Findings indicated that ARIMA achieved higher short-term forecasting accuracy compared to other models, validating its effectiveness for real-time disease monitoring.

Conceptual Framework

The development process begins with gathering relevant patient health records and location data, which are essential for monitoring and managing diseases. These collected data help define the system's purpose and inform the structure of its components. The system is composed of four main modules: The Account Module, Patient Module, Disease Tracking Module, Forecasting Module and Reporting Module. These core modules outline how the system handles user access, stores patient details, monitors the spread of diseases, predicts future outbreak and presents data through reports. The software development process includes an IDE, backend, frontend, and database. The Agile Scrum methodology was used to manage the development process from the initial concepts to the final output of the system. The project also integrates an ARIMA-based forecasting model and analytical tools to analyze disease case data and predict short-term case trends and fluctuations. Data security measures are applied to protect sensitive health information during processing and output. The final system is evaluated using ISO 25010 to ensure software quality and the Technology Acceptance Model (TAM) to assess user acceptance and usability.

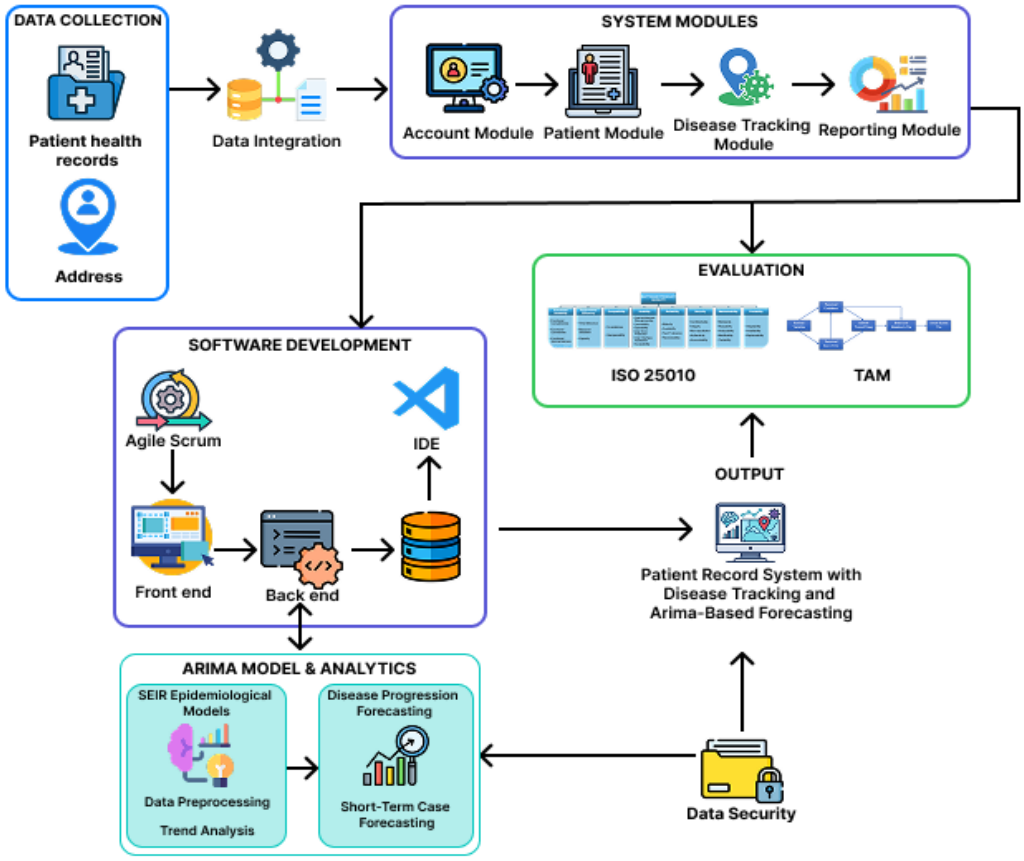


Figure 1. Conceptual Framework of the Patient Record System with Tracking and Forecasting for Top 5 Diseases

CHAPTER III

METHODOLOGY

Project Design

The design of the system was based on the prevailing architectural practices in order to address the rural health workers' needs for efficient management of patient data, surveillance of disease, and forecasting of disease trends.

The system architecture shown in Figure 2, consists of several key modules. The Account Module has login, user access, profile management, and logout. The module is secure in the way that only authorized personnel can access sensitive patient and system information. The Patient Module allows health providers to register patients, edit records, and access medical history, settings and accessible patient data management. The Disease Tracking Module enables tracking of cases, status, and case location mapping. This module is an important contributor to tracing the spread of communicable diseases within specific areas. The Reporting Module enhances data transparency and decision-making ability through report generation, export, viewing dashboard summaries, and system alerts retrieval. The ARIMA-based Analytics Module provides features for disease forecasting, risk assessment, and outcome-based updates. The features allow public health administrators to analyze historical case data, identify trends, and predict short-term changes in disease patterns, helping them respond more effectively to potential outbreaks.

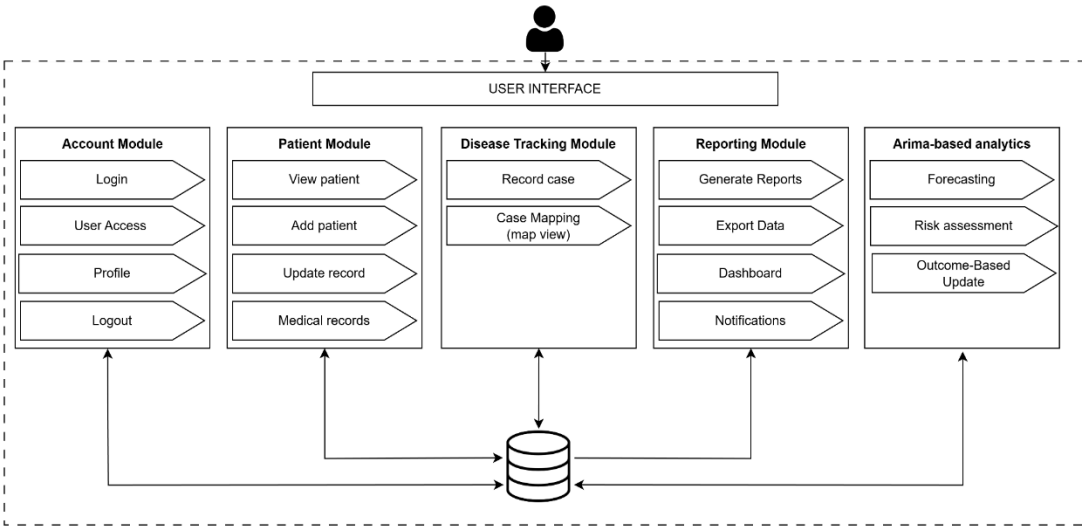


Figure 2. System Architecture

Software Development Model

Figure 3 illustrates the Agile Scrum software development cycle utilized in the Patient Record System with Disease Tracking and Forecasting development. Agile Scrum was utilized by the team, to have continuous interaction with users and continuous improvement. The Agile Scrum process divided the project into 1–4-week iterations, in which every sprint was intended to deliver some of the functional aspects of the system. Product backlog, sprint planning, sprint backlog, daily scrums, and delivering a finished product.

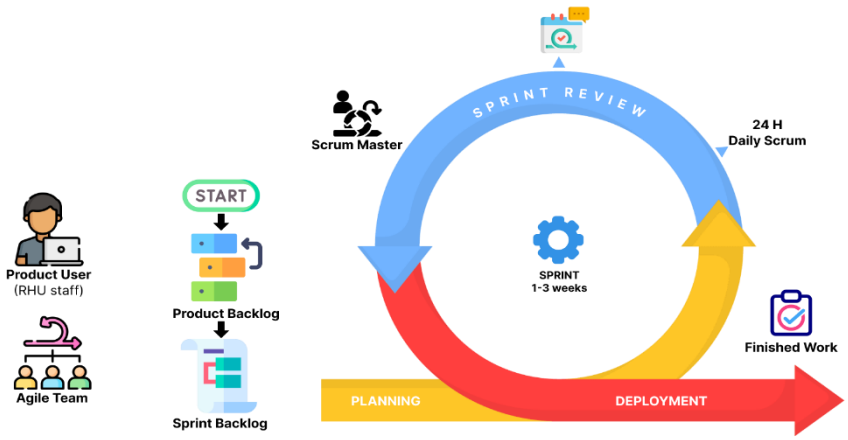


Figure 3. Agile Scrum

Product Backlog

Table 2 shows the product backlog for the admin section, outlining the key functionalities required for system management and oversight. These tasks serve as the foundational guide during the early stages of development, reflecting the client's essential requirements. The list includes core responsibilities such as logging into the system, managing user accounts and access permissions, updating patient records, overseeing disease tracking, and generating analytical reports. The estimated hours help in gauging the development workload and support resource allocation. By prioritizing tasks based on criticality, the backlog ensures that administrative functionalities are implemented effectively and align with the overall goals of the system.

Table 2. Product Backlog (Admin)

Priority	USER AS ADMIN Story	Est. hr/day
1	As an admin, I want to log in to the system securely to manage system access.	2hrs
2	As an admin, I want to manage staff accounts and set role-based access permissions.	3hrs
3	As an admin, I want to view patient records and monitor activity logs.	3hrs
4	As an admin, I want to access analytics for disease forecasting and system usage.	3hrs
5	As an admin, I want to export reports for documentation and compliance.	2hrs

Table 3 illustrates the product backlog for users, specifically the health staff. This section focuses on tasks directly linked to the daily operations within the system, such as creating patient records, updating medical history, and generating reports. The backlog ensures a user-friendly experience for health workers while maintaining the accuracy and integrity of health data. The prioritization and time estimates facilitate smooth implementation and help

developers focus on enhancing staff productivity and patient care through the system.

Table 3. Product Backlog (User)

Priority	USER AS STAFF Story	Est. hr/day
1	As a staff member, I want to log in and access my account with the appropriate permission.	1hr
2	As a staff member, I want to add and update patient records accurately.	3hrs
3	As a staff member, I want to record disease cases and update patient status.	3hrs
4	As a staff member, I want to view disease spread geographically using GIS maps.	2hrs
5	As a staff member, I want to generate and review summary reports for health monitoring.	3hrs

Sprint Planning & Sprint Backlog

Table 4 provides the sprint backlog, which defines the future functionality to be developed by the team. Items were taken from the product backlog and further elaborated into a more actionable sprint backlog, including specific tasks to be developed within a specific sprint cycle. Each task is linked to a specific user story and contains the estimated number of hours per day needed for completion. Development of each module is an organized process involving planning, coding, designing the user interface, and testing and debugging. Planning is the activity of defining the technical strategies and workflows of the system. The coding activity is targeted at writing the logic for the functionalities of the system. Designing the user interface is targeted at ease of use and compliance with system requirements. Testing and debugging are carried out to identify and correct any errors before deployment. This sprint backlog prioritizes must-have administrative functionalities such as login and authentication, staff account creation and administration, monitoring and

updating patient information, accessing analytical dashboards for disease summaries and forecasts, and system settings and export management. These features are prioritized to ensure the system accommodates the administrative functionalities needed for effective management of health data and disease tracking.

Table 4. Sprint Backlog (Admin)

Priority	Story	Task	Est. hr/day
1	As an admin, I want to log in to the system.	Design and develop a secure login page for admin with authentication.	2hrs
2	As an admin, I want to create and manage staff accounts.	Implement a user management interface to add, edit, and deactivate staff accounts.	3hrs
3	As an admin, I want to monitor and manage patient records.	Develop tools for viewing and updating patient data, including medical history and location.	3hrs
4	As an admin, I want to view disease case summaries and forecasts.	Create reporting dashboards that show real-time case summaries and ML-based predictions.	3hrs
5	As an admin, I want to manage system-wide settings and data exports.	Build settings module and data export functionality for reports and GIS outputs.	2hrs

Table 5 is a list of the current sprint backlog with an emphasis on user or staff-initiated activities. This sprint backlog is a prioritized list of development items within the sprint, prioritizing the highest-priority user requirements. Each item is a separate user story representing prominent system interactions from the point of view of a rural health staffer. The activities listed under each user story determine the development work required to implement that feature, along with the estimated number of hours per day to do so. The backlog emphasizes

key user activities like secure login, patient record additions and updates, disease case recording, and GIS-based access to maps for visualizing disease spread. Prioritizing these user-initiated features, the sprint seeks to improve system usability and facilitate day-to-day operations of health workers. The presence of spatial visualization tools demonstrates the mission of the system to empower users to respond more effectively and proactively to diseases.

Table 5. Sprint Backlog (Users/Staff)

Priority	Story	Task	Est. hr/day
1	As a user, I want to log in to the system.	Design staff login functionality with proper role-based access.	1hr
2	As a user, I want to register and update patient records.	Build forms and validation for adding and editing patient information.	3hrs
3	As a user, I want to record disease case updates.	Implement a module to input case status and updates.	3hrs
4	As a user, I want to generate reports of disease cases and trends.	Develop functionality for generating printable summary reports of case data.	3hrs
5	As a user, I want to view geographic spread and trends.	Provide access to GIS-based visualizations of current disease distribution.	3hrs

System Flowchart

Figure 4 shows the flow of actions for both staff and admin users within the system. The process starts by identifying the user role. If the user is a staff member, the system first checks whether they already have an active account. If not, they are instructed to contact the admin. Otherwise, they proceed to the login page. Upon successful login, staff are directed to the staff dashboard, where they can manage patient records through the Patient Module, track diseases, view reports and analytics, and access the map for visualizing

disease spread filtered by disease type. On the admin side, the process begins with a login prompt. Once authenticated, admins gain access to the admin dashboard, which includes additional capabilities such as staff account management and system settings. Admins can also perform key tasks such as managing patient records, monitoring disease tracking, generating reports and analytics, managing the ARIMA-based forecasting model for communicable diseases, accessing the GIS-based map feature, and printing reports while managing system configurations. The flowchart clearly outlines the sequence of operations and available features based on user roles. Both admin and staff users end their session by logging out, ensuring secure access and exit from the system. This structure promotes efficient task execution while maintaining a role-based separation of functionalities.

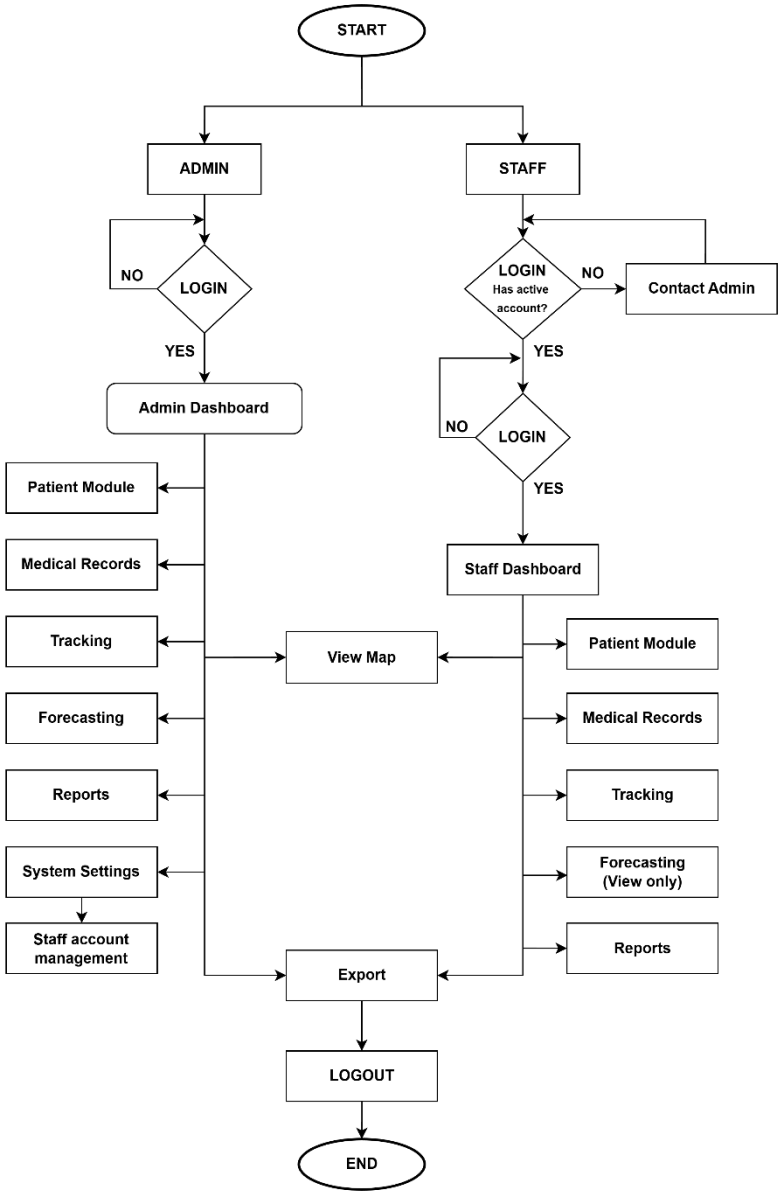


Figure 4. System Flowchart

Use Case Diagram

Figure 5 below illustrates the interactions between the two primary system actors: Admin and Staff. The admin has full control over the system functionalities, including account management, system settings, and ARIMA-based forecasting configuration. On the other hand, staff can access essential modules such as the Patient Module, Disease Tracking Module, and ARIMA-based forecasting as view only. Both users can generate reports and visualize

disease spread maps to assist in monitoring and decision-making. The diagram emphasizes role-based access and the collaborative nature of the system in supporting disease tracking and forecasting.

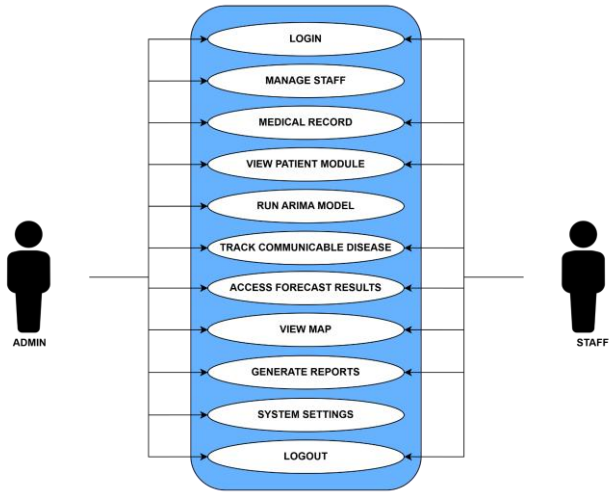


Figure 5. Use Case Diagram

Geographic Information System

The project incorporates a Geographic Information System (GIS) to visually map the distribution and spread of diseases within the locality. This feature is accessible to both the admin and staff users, enabling them to view reported disease cases across different geographic locations. The GIS map includes a drop-down filter to allow users to select specific types of risk level for monitoring. Once a risk level, the system highlights affected areas, helping health workers track and identify patterns in disease transmission. The GIS functionality aims to enhance decision-making and resource allocation, particularly in rural health units where rapid response to disease spread is critical. This component also support the visualization of data generated from the ARIMA-based forecasting model and reported cases.

Software Tools

MySQL. Serves as the primary relational database of the system. Through XAMPP and PhpMyAdmin, it stores structured data such as patient information, disease records, and forecasting results. MySQL ensures data integrity and supports efficient retrieval and management of records for monitoring and analysis.

PHP. Acts as the server-side scripting language responsible for handling database communication and backend logic. It processes requests from the frontend, performs CRUD (Create, Read, Update, Delete) operations on the database, and ensures secure data interaction between users and stored information.

Node.js. Operates as the runtime environment for executing backend JavaScript code. It is used for handling asynchronous operations, API calls, and real-time functionalities within the system. Node.js also bridges communication between the frontend and backend modules efficiently.

Vite React.js. Serves as the primary frontend framework for building the system's user interface. Using Vite for fast development and build optimization, React.js enables the creation of responsive and interactive components such as patient dashboards, disease tracking views, and forecasting displays.

Python. Is utilized to implement and execute the ARIMA-based forecasting model, which predicts possible trends of recorded diseases using historical case data. Python libraries such as pandas, NumPy, scikitlearn and stats models are used for data preprocessing, computation, and visualization, generating time-series forecasts integrated into the system's dashboard.

Tailwind CSS. Controls the visual layout, color, and typography of the system's interface. It provides a modern and responsive design framework that ensures a consistent and intuitive user experience across various devices and screen sizes.

HTML (Hypertext Markup Language). Defines the structure and layout of all web pages in the system, including login pages, dashboards, forms, and reports. It establishes the base for content presentation and navigation throughout the system.

Visual Studio Code. Serves as the integrated development environment (IDE) used for developing and testing the entire system. With built-in extensions for React, Node.js, PHP, and Python, it supports efficient coding, debugging, and collaboration throughout the development process.

LARAGON. Provides the local development environment that integrates Apache, PHP, and MySQL. It is used for hosting the backend locally and managing the database server during system testing and deployment preparation.

Hardware

Table 6 shows the minimum hardware required for development. 4GB RAM is the minimum required for smooth software performance. A quad-core processor such as Intel i3 or greater is sufficient for development. Windows 10 is the recommended OS for developing tool compatibility. One to two computers were used by the researchers for development.

Table 6. Hardware Specialization

HARDWARE	SPECIFICATION
RAM	4GB
CPU	Intel Core i3
OS	Windows 10

Table 7 shows the minimum requirements for PCs or Laptops to run the system effectively. A minimum of 4GB RAM and at least 2GB of free hard disk space is needed. Windows 10 is suggested due to compatibility with various applications and development tools. Access to a browser and a stable internet connection is also essential for utilizing all system functionalities.

Table 7. Minimum Requirements for PC or Laptop

MINIMUM REQUIREMENTS FOR PC OR LAPTOP	SPECIFICATION
RAM	4GB
PROCESSOR	Intel Celeron
OS	Windows 10
HARD DISK	At least 2GB free space

Table 8 outlines the minimum requirements for smartphones or tablets to access the system efficiently. Android 10 or older versions are recommended to avoid compatibility issues. A browser and internet connection are necessary for accessing features such as tracking and reports.

Table 8. Minimum Requirements for Smartphone or Tablet

MINIMUM REQUIREMENTS FOR SMARTPHONE OR TABLET	SPECIFICATION
RAM	4GB
PROCESSOR	Quad-core
ANDROID OS	Android 10 and older
BROWSER	At least 2GB free space

Project Testing and Evaluation

The development team conducted a series of tests to ensure that the Patient Record System with Disease Tracking and ARIMA-Based Forecasting for Top 5 Diseases performs correctly and meets the required system specifications. The testing focused on verifying that each module works properly on its own and that all components function together without issues. User feedback was also gathered to assess the system's usability, reliability, and overall performance based on the ISO 25010 Software Product Quality Model. The results of these tests were used to make necessary refinements before final deployment.

Software Testing

Figure 6 presents the steps of software testing, showing how individual software units and components are evaluated. Its objective is to ensure that all parts of the software are functioning as they should and according to what has been designed. This figure describes different testing levels, which are used to check whether components of the system have been made to function. Each unit is examined based on its expected behavior. A unit is defined as the smallest testable portion of software, typically consisting of one or more inputs and a single output. In procedural programming, a unit may refer to an individual function, program, or method, while in object-oriented programming, the smallest unit is usually a method contained within a class whether abstract, base, or derived. Unit Testing (UT) provides several benefits, including increased developer confidence during both development and maintenance. In this study, the researchers decomposed the system into smaller components

and tested each one independently to determine whether every component performed as intended.

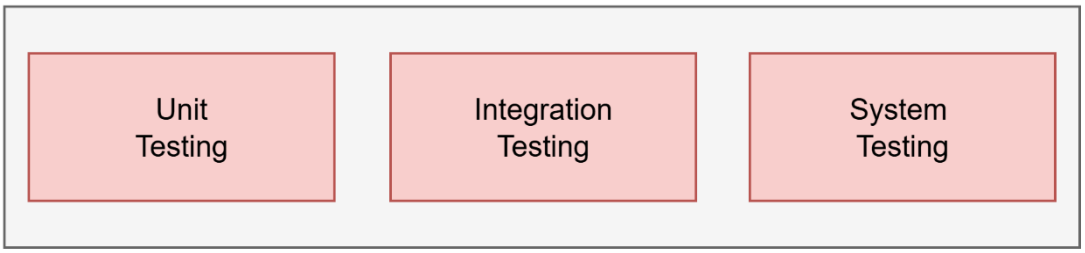


Figure 6. Stages of Testing

Unit Testing

Figure 7 illustrates the unit testing phase. This testing was performed to verify that each individual module, such as the patient record module, disease tracking module, and the forecasting module operates correctly on its own. Each unit was tested separately to ensure that its functions perform as expected and produce accurate results. The purpose of this phase was to detect and resolve errors early, ensuring that each module is stable before integration.

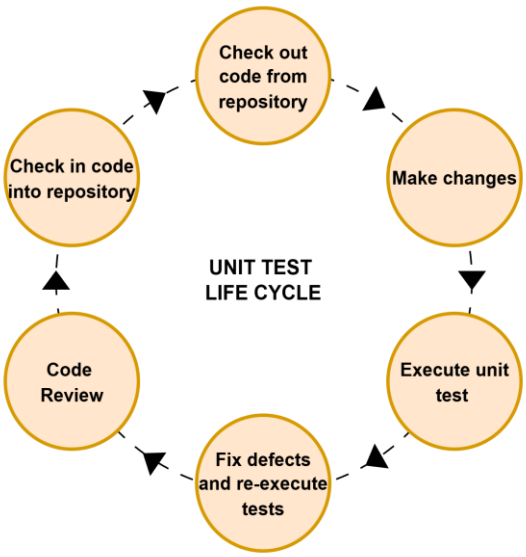


Figure 7. Unit Testing

Integration Testing

Figure 7 presents the integration testing phase, where all modules of the system were combined and tested together to ensure seamless operation. This testing verified that the patient record module, disease tracking features, and forecasting component work properly when integrated. It also ensured that data flows smoothly between modules and that the system performs consistently based on the expected workflow. The feedback gathered during this phase helped identify minor adjustments and confirmed that the system is functioning correctly and ready for real-world use.

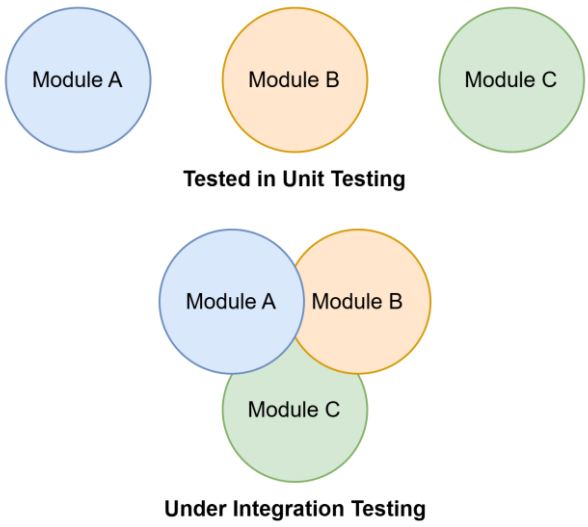


Figure 8. Integration Testing

Evaluation Matrix

The system evaluation for the Patient Record System with Patient Tracking and ARIMA-Based Forecasting for Top 5 Diseases in Los Banos, Laguna is based on the ISO 25010 quality standards. ISO 25010 provides a structured quality model composed of eight attributes used to assess both the functional and non-functional aspects of a software system. To evaluate the system to be developed, the developers do the beta tests for the I.T experts.

The evaluation covers internal and external system designs, structure, and overall functionality. is answerable by five responses which is 5 (Excellent) – 4.20 to 5.00 4 (Good) – 3.40 to 4.19, 3 (Fair) – 2.60 to 3.39, 2 (Needs Improvement) – 1.80 to 2.59, 1 (Poor) – 1.00 to 1.79. Each system module and feature will be individually tested using this standard. Continuous testing during the development phase allows early detection and resolution of faults, ensuring that the final output is a high-quality web-based system ready for deployment.



Figure 9. Quality Characteristic of ISO 25010

Technology Acceptance Model (TAM)

The Technology Acceptance Model (TAM) is used to assess how users perceive and interact with the Patient Record System with Disease Tracking and ARIMA-Based Forecasting for Top 5 Diseases. TAM evaluates the system's performance, functionality, and overall usefulness to users by collecting feedback using surveys to assess categories like Perceived Ease of Use, Perceived Usefulness, Attitude Toward Using, and more. The responses were ranging from 1 (Not Acceptable) to 4 (Highly Acceptable).

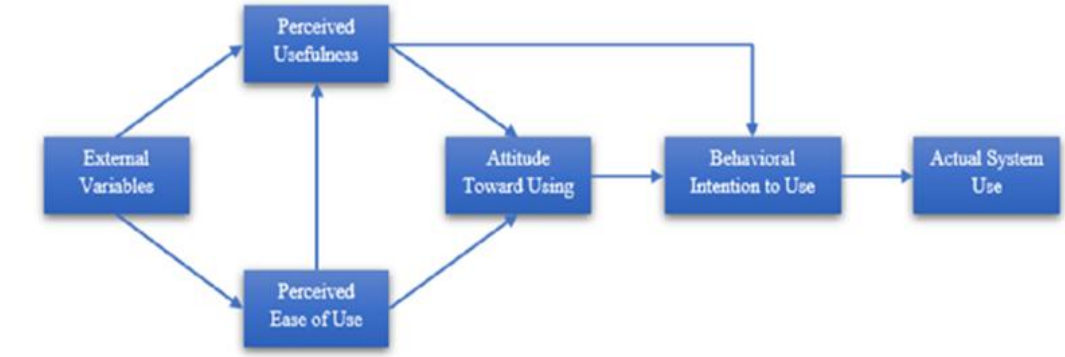


Figure 10. Technology Acceptance Model (TAM)

Model Evaluation

The performance of the ARIMA-based forecasting model was evaluated using standard accuracy metrics to ensure the reliability and precision of its predictions. Mean Absolute Error (MAE) and Root Mean Square Error (RMSE) were employed to quantify how closely the model's forecasts align with the actual recorded data. These metrics provide insight into the model's predictive accuracy, helping determine its effectiveness in anticipating trends and potential fluctuations in disease cases. A lower value in both MAE and RMSE indicates a more accurate model, thereby supporting informed decision-making and planning for preventive measures and health interventions.

Mean Absolute Error (MAE)

Mean Absolute Error (MAE) measures how close the predicted values from the ARIMA model are to the actual recorded values. It calculates the average absolute difference between the forecasted and actual case counts, indicating the model's prediction accuracy. A lower MAE value signifies higher forecasting precision.

The formula used was:

$$MAE = \frac{1}{n}\sum_{i=1}^{n} |\, y_i - \hat{y}_i|$$

Where:

- n = number of observations
- y_i = predicted value from the ARIMA model
- $\hat{y}_i$ = real observed value

Root Mean Square Error (RMSE)

Root Mean Square Error (RMSE) was also used to further assess the accuracy of the ARIMA-based forecasting model. RMSE measures the square root of the average squared differences between the predicted values and the actual recorded cases. Unlike MAE, which treats all errors equally, RMSE gives a higher weight to larger errors, making it useful in identifying how well the model performs when sudden increases or decreases in cases occur. A lower RMSE indicates better forecasting performance and a closer fit between the model's predictions and the actual disease data.

The formula used was:

$$RMSE = \sqrt{\frac{1}{n}\sum_{i=1}^{n} (y_i - \hat{y}_i)^2}$$

Where:

- n = number of observations
- y_i = predicted value from the ARIMA model
- $\hat{y}_i$ = actual recorded value

Dataset

The dataset used in this study was obtained from the Rural Health Unit and consists of aggregated disease case data covering the years 2021 to 2024. It includes real attributes such as the barangay where the cases were reported, the age group and sex of the affected individuals, the type of disease, and the total number of reported cases per barangay for each year. A total of one thousand case entries were extracted from the aggregated data and used to populate the system. Due to privacy restrictions under the Data Privacy Act of 2012, the RHU does not release patient-level identifiers such as names, exact addresses, or consultation histories. To allow testing of system functionalities that require individual patient entries including record creation, retrieval, and updating, the researchers generated synthetic patient-level data. These synthetic records consist of fictional names, identification numbers, and consultation timestamps, and they were carefully constructed to reflect the distributions and patterns present in the real aggregated data. The ARIMA-based forecasting model, which predicts trends for the for the top five recorded diseases in the rural health unit, used only the real aggregated case counts. The synthetic patient data was employed solely for functional testing of the system and did not influence the predictive outputs.

Data Preprocessing

Before the dataset was used in the system and for ARIMA-based forecasting, it underwent a preprocessing phase to ensure consistency, accuracy, and usability. The aggregated data from the Rural Health Unit was first reviewed to identify and handle missing or inconsistent entries. All records were verified to ensure that the total number of cases per barangay,

age group, and sex matched the reported counts. The data was then organized in a structured format suitable for input into the system and the forecasting model. Train-Test Split: To evaluate the forecasting model's performance and prevent overfitting, the pre-processed time-series data was split into training and testing sets using an 80/20 ratio. The first 80% of the chronological data was used to train the ARIMA model, allowing it to learn the underlying patterns, trends, and seasonality in disease occurrences. The remaining 20% of the data was reserved as the test set to validate the model's predictive accuracy on unseen data. This split approach ensures that the model's performance metrics reflect its ability to generalize to future disease trends rather than merely fitting historical data. For the synthetic patient-level data, pre-processing involved generating fictional names, identification numbers, and consultation timestamps while maintaining the patterns and distributions observed in the real aggregated data. This step ensured that the system modules for patient records could function correctly without compromising privacy. Finally, the dataset was formatted to align with the requirements of the ARIMA model, including proper time-series ordering and aggregation by month and disease type. This preprocessing process ensured that both the real and synthetic datasets were clean, complete, and suitable for accurate forecasting and functional testing of the system.

Model Development

The development of the forecasting model began with the selection of the ARIMA (Autoregressive Integrated Moving Average) approach, which is widely used for time-series analysis and prediction of trends in recorded diseases. The model was built using the real aggregated case counts provided

by the Rural Health Unit, covering the years 2021 to 2024. The data was first structured into monthly intervals for each disease type to create a continuous time series.

Parameter Selection and Model Training for the ARIMA model, including the autoregressive order (p), differencing order (d), and moving average order (q), were determined through systematic analysis of the training data patterns. This involved examining autocorrelation function (ACF) and partial autocorrelation function (PACF) plots to identify appropriate lag orders and stationarity requirements. The model was trained to recognize historical trends and patterns in disease occurrence across different barangays. Multiple ARIMA configurations were tested to identify the optimal parameter combination that minimized information criteria (AIC/BIC) on the training set. Once the model was fitted on the training data, its predictive performance was evaluated using the reserved test set. The model's forecasts for the test period were compared against actual recorded case counts to calculate accuracy metrics such as Mean Absolute Error (MAE) and Root Mean Square Error (RMSE). This validation approach ensured that the reported model performance reflects the ability to predict future trends rather than merely fitting historical data. The ARIMA-based forecasting model provides estimates for future case counts while relying exclusively on real aggregated data, ensuring that the predictions reflect actual historical trends observed in the community. It is important to note that the synthetic patient-level data generated for system testing was not used in the model development or validation process. This separation ensures that the forecasting results remained valid and based solely on authentic

epidemiological data from the Rural Health Unit, maintaining the scientific integrity and reliability of the predictive model.

Population of the Study

Table 9 presents the respondents, which included three (3) Information Technology (IT) experts. They were selected to assess the Patient Record System with Disease Tracking and ARIMA-Based Forecasting for Top 5 Diseases. The IT experts assessed the system according to the ISO 25010 software quality model, that is, functionality, usability, reliability, and efficiency. Their technical expertise ensured that the assessment results were accurate and industry-aligned.

Table 9. Respondents of the Research Project (ISO 25010)

CATEGORY OF RESPONDENTS	SAMPLE
I.T expert	3
Total	3

Table 10 shows the respondents selected for the project. The respondents are the admin, who operates the system, and the RHU staff members, who actively utilize the system. These individuals were selected as they are the primary users of the Patient Record System with Disease Tracking and ARIMA-Based Forecasting. The number of respondents is seven (7) Admin and ninety-three (93) Rural Health Unit and local health staff members. Their feedback is essential in evaluating the system’s usability, functionality, and acceptance based on the Technology Acceptance Model (TAM).

Table 10. Respondents of the Research Project (TAM)

CATEGORY OF RESPONDENTS	SAMPLE
Staff	93
Admin	7
Total	100

Sampling Design

The study utilized purposive sampling, specifically selecting respondents who are directly involved in patient record management and disease monitoring at the Rural Health Unit (RHU). The respondents include system users such as health staff and the system administrator who are knowledgeable in healthcare operations. A total of 100 respondents, 93 staff and 7 admin participated in the evaluation based on the Technology Acceptance Model (TAM).

Data Collection Instrument

The data were collected by conducting interviews and providing survey questionnaires to the respondents. The gathered data served as the basis for developing the Patient Record System with Disease Tracking and ARIMA-Based Forecasting for Top 5 Diseases.

The primary tool used for data gathering was a structured survey questionnaire. The instrument was divided into two sections. The first set is the demographic profile of the respondents and the second set of the system evaluation form based on ISO 25010 software quality standards, such as functionality, reliability, usability, efficiency, compatibility, security, maintainability and portability. Respondents rated each item using Likert scale ranging from 1 ("Poor") to 5 ("Excellent"). Open-ended questions were also included to gather qualitative feedback on system usability and for improvement.

Statistic Treatment

The study used descriptive statistics to analyze the collected data. The Weighted Mean was computed to determine the overall perception of the respondents towards the developed system.

The formula used was:

$$Weighted\ Mean = \frac{\sum(x_i \cdot w_i)}{\sum w_i}$$

Where:

- $\sum$ = sum of all responses
- x_i = individual scores or responses
- w_i = total number of respondents

Interpretation of the weighted mean results followed this scale:

Table 11. Weighted Interpretation

Rating	Scale Range	Verbal Interpretation
5	4.21 – 5.00	Strongly Agree
4	3.41 – 4.20	Agree
3	2.61 – 3.40	Neutral
2	1.81 – 2.60	Disagree
1	1.00 – 1.80	Strongly Disagree

Standard Deviation (SD)

The standard deviation was used to measure the spread of respondents' answers around the calculated mean. A lower standard deviation indicates that respondents' ratings were closely clustered around the mean, reflecting similar perceptions among participants, while a higher standard deviation suggests

more diverse opinions. The formula for calculating the sample standard deviation is:

$$SD = \sqrt{\frac{\sum(x - \underline{x})^2}{n - 1}}$$

Where S represents the sample standard deviation, X is each individual response in the sample, $\bar{X}$ is the mean of the responses, and n is the total number of responses in the sample.

Ethical Consideration

This study strictly follows ethical guidelines to ensure the protection, privacy, and rights of all participants and the confidentiality of data used. The researchers first submitted a formal request letter to the Office of the Mayor to seek approval for conducting the study in the community. After receiving the mayor's approval, permission was formally requested from the Rural Health Unit, including the heads and staff, to conduct interviews and collect data. A consent letter, signed by the research adviser, was also provided to ensure transparency and secure approval for participation.

Participants were fully informed about the purpose of the study, the nature of the questions, and their right to refuse participation or withdraw at any time without any negative consequences. Participation was entirely voluntary, and all information collected from participants was treated with strict confidentiality and used solely for academic purposes related to the development and evaluation of the Patient Record System with Disease Tracking and ARIMA-based Forecasting for for the top five recorded diseases.

The dataset provided by the RHU contained only aggregated disease case information, including barangay, age group, sex, disease type, and case counts. No patient-level identifiers such as names, addresses, or medical histories were provided due to data privacy restrictions under the Data Privacy Act of 2012. To enable system testing for modules requiring patient-level entries, the researchers generated synthetic patient data consisting of fictional names, identification numbers, and consultation logs. These synthetic records do not correspond to any real individuals and were used exclusively to test system functionality. This approach ensured that the study adhered to ethical standards, protected the privacy of the community, and complied with all applicable legal and institutional guidelines.

CHAPTER IV

RESULTS AND DISCUSSION

This chapter presents the results and discussion based on the objectives of the study. The developed Patient Record Management System with Disease Tracking and ARIMA Forecasting was evaluated in terms of its ability to support Rural Health Units (RHUs) in managing patient information, monitoring communicable diseases, and generating data-driven predictions for public health planning. The following sections discuss how the system fulfills each research objective, supported by the functionalities implemented in each module.

Effectiveness of the secure patient record system that ensures efficient recording, storage, and retrieval of patient health records.

Before accessing any part of the system, users must log in through a secure authentication page. The login module ensures that only authorized RHU personnel such as administrators and staff can enter the system. User roles determine the level of access, helping maintain data confidentiality and preventing unauthorized viewing or modification of sensitive patient information. This feature strengthens the integrity and privacy of all stored records. Figure 11 shows the login interface of the system.

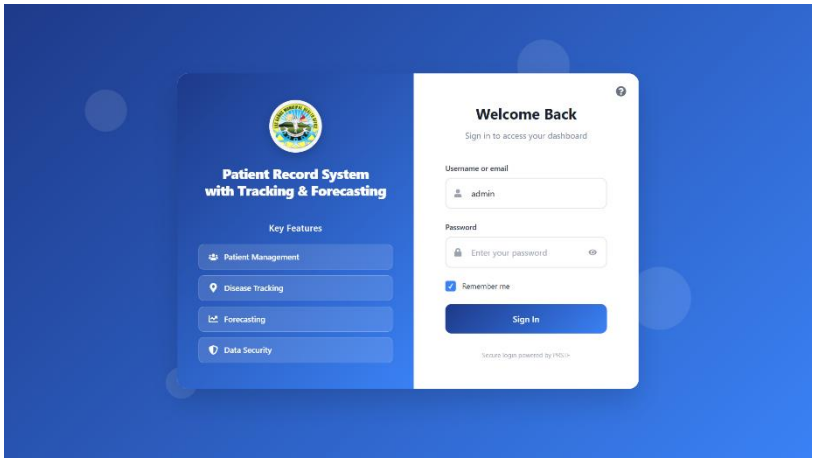


Figure 11. User Interface for System Login

The system features a dedicated Patient Module that consolidates all registered patients in a searchable and sortable table. Staff can easily view essential patient details and access their corresponding records. This interface provides an organized overview of all stored patient information, making retrieval faster and reducing reliance on manual documentation. Figure 12 shows the system's patient list interface where records are displayed in a structured format for convenient navigation.

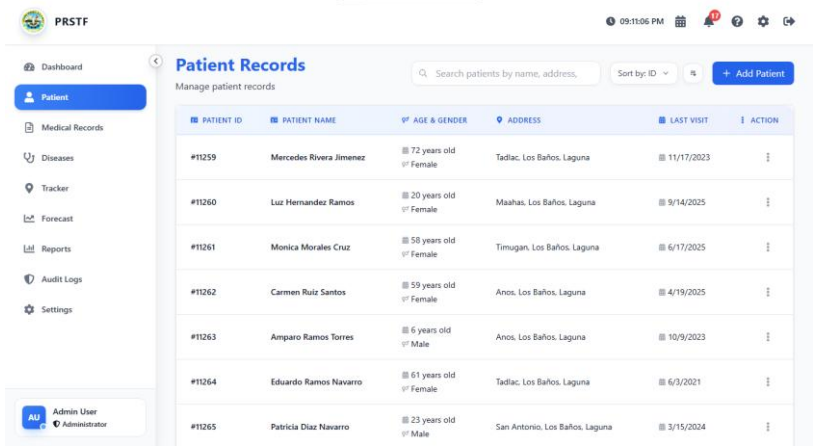


Figure 12. User Interface for Patient List

When adding a new patient, the system provides a guided form that clearly indicates required information, ensuring that all essential details are properly encoded. Optional fields are also available for additional patient data

when needed. This structure helps maintain consistency across all patient entries while minimizing the possibility of incomplete records. Figure 13 shows the interface used for registering new patient information.

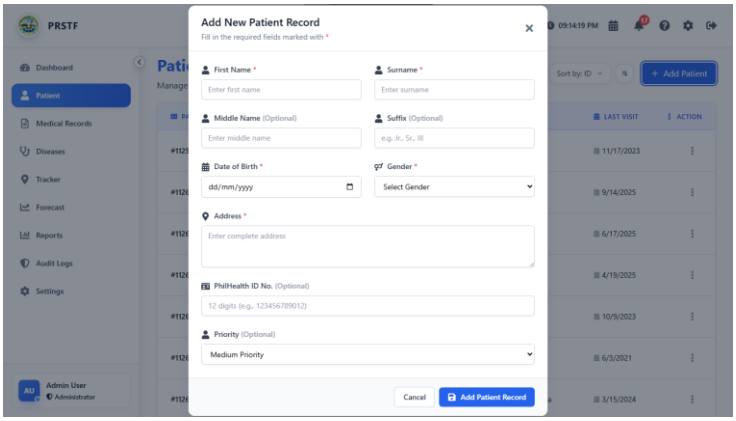


Figure 13. User Interface for Adding Patient Records

In addition to basic information, the system includes a Medical Records module where staff can view diagnosis summaries, recent visits, and other clinical details. The records are presented in a structured list that allows health workers to quickly identify patients needing follow-up or further assessment. Figure 14 illustrates the interface showing the list of medical records stored in the system.

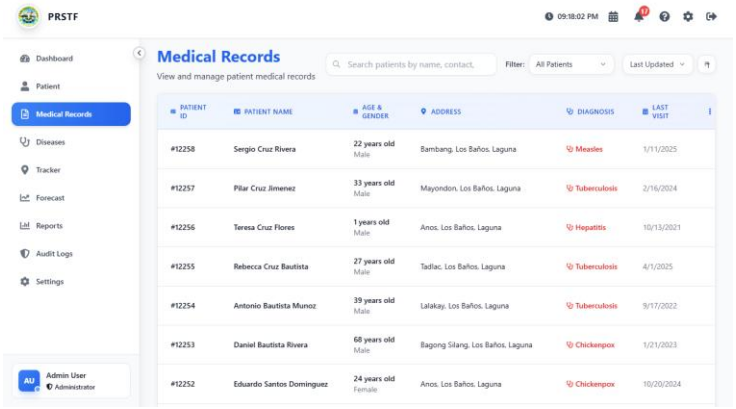


Figure 14. User Interface for Patient Medical Records List

Selecting a patient opens a detailed medical profile that contains demographic information, clinical notes, vital signs, and a complete

consultation history. This section serves as a centralized record of each patient's medical background, supporting continuity of care and informed clinical decision-making. Health personnel may update assessments or add new entries as needed. Figure 15 shows the patient medical profile and consultation history interface.

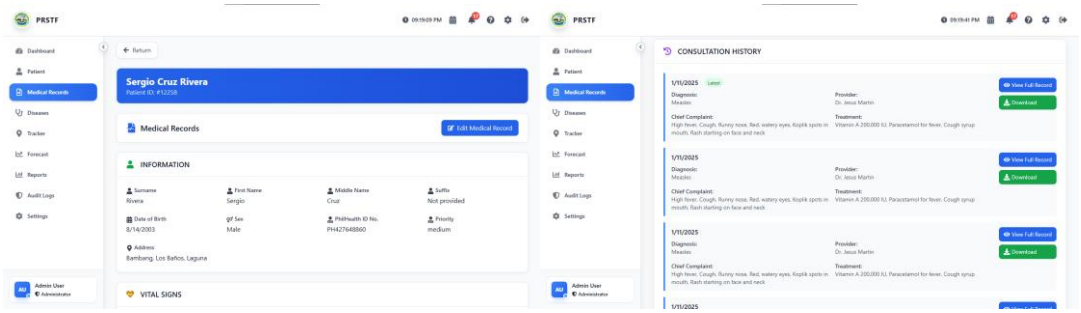


Figure 15. Patient Medical Record and Consultation History Interface

These features provide RHU staff with a secure and organized means of managing patient data, ensuring efficient storage, retrieval, and updating of health records.

Effectiveness of the map-based disease tracking system for visualizing case distribution across barangays.

To support disease surveillance, the system includes a Disease Tracking module that displays communicable disease case distribution within Los Baños. The top section of this module summarizes key indicators such as the number of monitored barangays, total recorded patients, and areas classified as high risk. This summary provides health workers with a quick overview of the current disease situation. Figure 16 presents the interface showing the disease tracking.

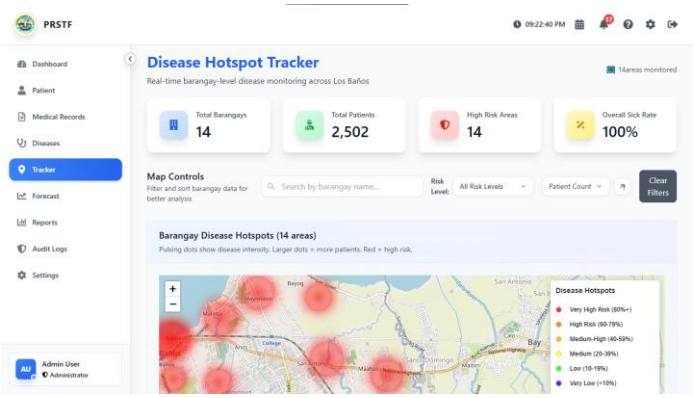


Figure 16. Disease Tracking Interface

Users may filter case data by disease, barangay, or risk level using the toolbar provided. These controls help staff focus on specific diseases or locations and easily assess the severity of cases within the community. Below the toolbar, a map visualization displays color-coded indicators representing the number of cases reported in each barangay. High-risk areas are highlighted, allowing staff to immediately identify clusters of disease occurrence. Figure 17 shows the barangay-level hotspot map used in visualizing case distribution.

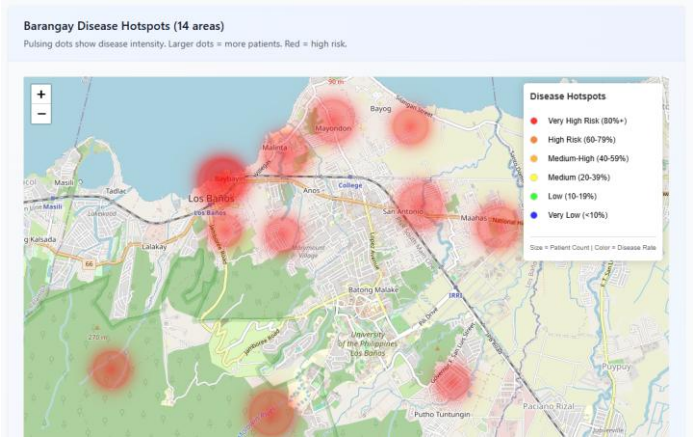


Figure 17. Barangay-Level Disease Hotspot Map

This spatial representation strengthens local disease monitoring efforts by enabling RHU personnel to observe case patterns, identify emerging hotspots, and support targeted public health interventions.

Short-term prediction of the top five diseases using ARIMA-Based Forecasting.

The Forecasting module allows users to generate short-term predictions for the top communicable diseases recorded in the system. Users can select forecast settings such as the type of forecast (overall or barangay-level), duration of prediction, and disease category when applicable. This configurable interface ensures flexibility depending on the needs of the RHU. Figure 18 display the forecasting parameter selection page.

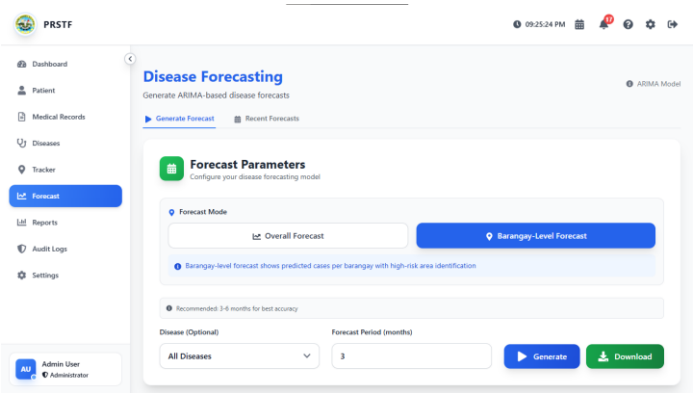


Figure 18. Forecast Parameter Configuration Interface

After generating results, the system provides a Forecast Summary presenting key metrics such as the number of diseases analyzed, length of forecast period, and the most recent case counts. Below the summary is an interpretation section that outlines expected trends, peak periods, and the disease predicted to have the highest case count. This section also includes recommendations to support planning and resource allocation. Figure 19 shows the forecast summary and interpretation interface.

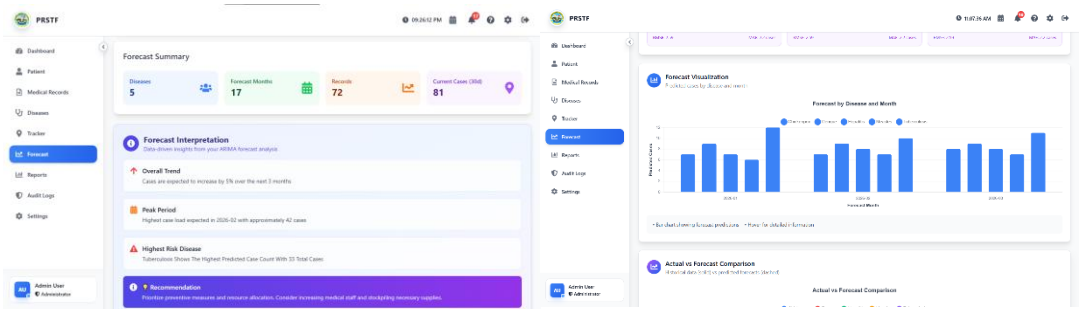


Figure 19. Forecast Summary and Interpretation Interface

The module also provides detailed prediction outputs for each disease, including projected monthly case values and error metrics such as RMSE and MAE. Visualizations such as bar charts and line graphs comparing actual and forecast data help staff analyze the model's accuracy and interpret the forecasted trends more easily. Figure 20 illustrates the graphical forecast visualization generated by the system.

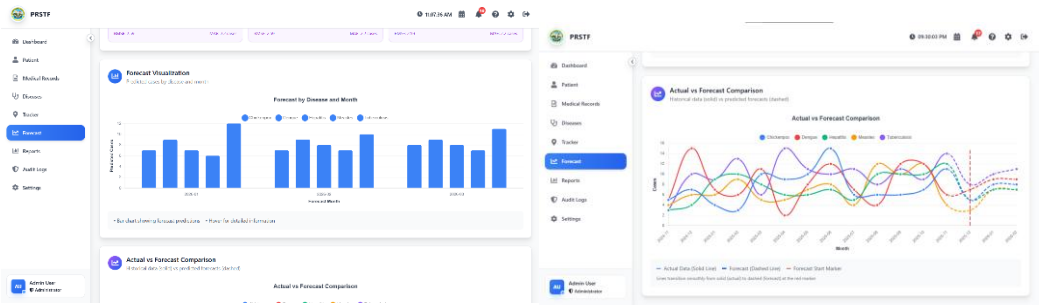


Figure 20. Forecast Visualization and Actual vs. Forecast Comparison

Barangay-level forecasting follows a similar structure but provides predictions for each barangay affected by the monitored diseases. This supports localized planning and allows RHUs to identify which areas may require additional attention in the upcoming months.

Results of the ISO 25010 quality characteristics and Technology Acceptance Model (TAM).

Project Testing and Debugging

A. Unit Testing

Unit testing was performed to verify that the essential features of the system functioned correctly in isolation. Each module was tested by simulating typical user actions to confirm that expected outputs were produced. The testing concentrated on the system's core functions, such as user authentication, patient management, medical records handling, disease tracking, and user account settings.

Table 12. Unit Testing Evaluation

Unit Test ID	Test Conducted	Expected Result	Actual Result	Remarks
1	Testing the user authentication module by attempting login using both valid and invalid inputs to confirm proper acceptance and rejection of credentials	System correctly processes valid inputs and blocks incorrect ones	Behavior matched expected results	Passed
2	Testing the dashboard module by loading statistics, alerts, and graphical summaries to verify the accurate display of system data	Dashboard displays updated and complete information without errors	Displayed smoothly	Passed
3	Testing the patient management functions by adding, editing, deleting, and searching patient information to confirm the system's ability to handle records	System performs all record operations correctly	All operations worked properly	Passed
4	Testing the medical records module by saving, updating, and retrieving examination details to ensure proper recording and retrieval	Examination data is saved and displayed properly	Worked as intended	Passed
5	Testing the disease	Updated data	Displayed	Passed

	tracking module by updating disease entries and checking if changes appear in summaries and lists	should appear instantly and accurately	accurately	
6	Testing the forecasting module by generating basic disease trend visualizations based on existing data	Forecast results appear based on available information	Output displayed successfully	Passed
7	Testing the reports module by generating basic summaries and downloadable outputs to verify that the system can compile and present data correctly	Reports should generate without errors and display complete information	Reports were produced successfully	Passed
8	Testing the user account management under the settings module by adding new users, updating account details, and deactivating existing accounts	User updates and status changes are saved and reflected correctly	Account changes applied successfully	Passed

B. Integration Testing

Integration testing was conducted to determine whether the different modules of the system worked correctly when combined. This phase focused on the most essential interactions between components to ensure that data remained consistent and that user actions in one module were reflected properly in another.

Table 13. Integration Testing Evaluation

Unit Test ID	Test Conducted	Expected Output	Actual Output	Remarks
1	Testing the connection from user authentication to the dashboard to confirm that successful login leads to the correct dashboard view	Dashboard loads with the appropriate user information	Loaded smoothly	Passed
2	Testing the link between patient management and medical records, ensuring that selecting a patient retrieves the correct examination data	Corresponding medical record displays correctly	Displayed accurately	Passed
3	Testing how changes made in patient records affect the dashboard, verifying that statistics update after modifications	Dashboard totals update based on changes	Updated correctly	Passed
4	Testing the impact of user roles set in the settings module on system access, confirming that access rights reflect different user types	System features adjust based on assigned role	Restrictions applied correctly	Passed

C. System Testing

The system testing phase served as the final assessment of the Patient Record System with Disease Tracking and ARIMA-Based Forecasting. This stage ensured that all modules worked smoothly together and that the system could operate reliably under real-world conditions. The testing focused on system stability, responsiveness, and how well data moved across modules once everything was combined.

During system testing, the Patient Module performed consistently even when handling multiple record updates, ensuring that patient information was saved and retrieved without delay. The Medical Records Module processed examination entries smoothly, showing that clinical data remained intact during repeated edits and retrievals. The Disease Tracking Module also worked reliably, displaying updates in maps and summaries without lag, allowing users to monitor case movements. The Forecasting Module generated projections based on existing datasets without errors, confirming that the ARIMA feature works properly when integrated with live system data. Reports were generated correctly as well, showing complete and organized summaries that RHU staff can use for planning.

The full end-to-end flow from login to data entry, tracking, forecasting, and report generation ran without interruptions. Data moved securely and consistently across modules, and the system responded well throughout the testing process. These results confirmed that the PRMS–ARIMA system is stable, reliable, and ready for continuous use in local health settings. The system testing verified that the platform supports the study’s goals of improving patient record management, strengthening disease monitoring, and offering useful short-term forecasts for decision-making.

Table 14. System Testing Evaluation

Unit Test ID	Test Conducted	Expected Results	Actual Results	Remarks
1	Test the Login Module under multiple simultaneous logins.	The system should handle multiple sessions without slowdown.	Logins processed smoothly with no errors.	Passed
2	Test Patient and Medical Records Modules during continuous add/edit/delete operations.	The system should perform all actions without losing or corrupting data.	All record processes worked correctly and consistently.	Passed
3	Test Disease Tracking Module for updates of case entries.	Updated cases should appear instantly on summaries and map view.	Updates displayed accurately.	Passed
4	Test Forecasting Module performance during repeated generation of ARIMA forecasts.	Forecasts should compute correctly and display without lag.	Forecasts generated successfully each time.	Passed
5	Test Reports Module for compiling and exporting summaries.	Reports should generate complete and error-free outputs.	Reports were generated accurately and consistently.	Passed

Forecast Accuracy Metrics for the ARIMA Model Result

Table 15 presents the forecast accuracy metrics generated by the ARIMA model for the top five diseases, with accuracy assessed using the Mean Absolute Error (MAE) and Root Mean Square Error (RMSE) following the formulas discussed in Chapter 3, showing how close the predicted cases are to the actual data, where lower MAE and RMSE indicate more accurate and stable forecasts.

Based on the results, Chickenpox, Dengue, Hepatitis, and Tuberculosis showed low error values, meaning the ARIMA model was able to follow their

trends closely and produce consistent short-term predictions, while Measles recorded the highest error values among the five diseases, suggesting that its case pattern is more irregular, making it harder for the model to forecast with the same level of accuracy, and overall, the findings show that the ARIMA model performs well for most diseases and can provide useful forecasts that help the Rural Health Unit in monitoring trends and preparing for possible increases in cases.

Table 15. Forecast Accuracy Metrics for the ARIMA Model

Disease	RMSE	MAE	Interpretation
Chickenpox	2.52	2.10	The model produced low forecast errors, showing reliable short-term predictions.
Dengue	2.62	2.00	Forecast accuracy is high, with minimal differences between actual and predicted values.
Hepatitis	2.81	2.60	Errors are small and acceptable, indicating that the model follows the actual trend well.
Measles	4.88	4.70	Error values are slightly higher due to its more variable pattern, but the model still captures the general movement of cases.
Tuberculosis	2.59	2.20	The model shows good accuracy, with predictions close to the actual case counts.

ISO 25010 Evaluation Results

The table 16 presents the post-evaluation results of the system based on the ISO 25010 software quality characteristics. Three (3) IT experts evaluated the system using a 5-point Likert scale, and the weighted mean for each characteristic was computed to determine the overall quality of the system. Based on the results, the system obtained an overall mean of 4.89,

which is interpreted as Excellent. All characteristics received high ratings, showing that the system performs well in terms of functionality, efficiency, usability, reliability, security, and maintainability. These findings indicate that the system meets the expected software quality standards and is suitable for deployment and continued use.

Table 16. ISO 25010 Evaluation Results

Characteristics	Mean	Standard Deviation	Interpretation
Functional Suitability	5.00	0.00	Excellent
Performance Efficiency	4.93	0.12	Excellent
Compatibility	4.67	0.47	Excellent
Usability	4.94	0.12	Excellent
Reliability	4.93	0.12	Excellent
Security	5.00	0.00	Excellent
Maintainability	4.73	0.25	Excellent
Flexibility	5.00	0.00	Excellent
Safety	4.80	0.26	Excellent
Composite Mean	**4.89**		**Excellent**

Legend: 1–1.80 = Poor, 1.81–2.60 = Fair, 2.61–3.40 = Good, 3.41–4.20 = Very Good, 4.21–5.00 = Excellent. n=3 IT experts

Technology Acceptance Model (TAM) Results

This section presents the acceptability of the system as evaluated through the Technology Acceptance Model (TAM). A total of 100 respondents assessed the system using a 4-point Likert scale across four areas: Perceived Usefulness, Perceived Ease of Use, Attitude Toward Using, and Behavioral Intention to Use. TAM provides insight into how users view the system in terms

of its practicality, usability, and their willingness to adopt it in actual health center operations.

Table 17 shows the respondents' assessment of Perceived Usefulness. The results reveal that the system received high ratings across all indicators. Respondents shared that the platform helps them accomplish tasks more efficiently, improves patient data management, and supports effective disease monitoring. The forecasting feature was also viewed as helpful for decision-making. With all weighted means falling under "Highly Acceptable," users recognize the system as a valuable tool in their daily responsibilities.

Table 17. Perceived Usefulness of the System

Indicator	Mean	Standard Deviation	Interpretation
Helps me complete work efficiently	**3.55**	**0.55**	Highly Acceptable
Improves management of patient data	**3.54**	**0.56**	Highly Acceptable
Forecasting helps decision-making	**3.52**	**0.59**	Highly Acceptable
Useful for monitoring disease cases	**3.58**	**0.53**	Highly Acceptable
Contributes to faster response and decisions	**3.55**	**0.56**	Highly Acceptable
Composite Mean	**3.55**		**Highly Acceptable**

Legend: 1.00–1.75 = Not Acceptable, 1.76–2.50 = Slightly Acceptable, 2.51–3.25 = Acceptable, 3.26–4.00 = Highly Acceptable.

Table 18 presents the respondents' ratings for Perceived Ease of Use. The consistently high scores indicate that users find the system simple to operate and easy to navigate. The interface was considered user-friendly,

functions were accessible, and tasks could be completed without significant technical guidance. These results show that users were able to learn and use the system comfortably.

Table 18. Perceived Ease of Use of the System

Indicator	Mean	Standard Deviation	Interpretation
System is easy to learn	**3.55**	**0.55**	Highly Acceptable
Interface is simple and user-friendly	**3.58**	**0.53**	Highly Acceptable
Tasks can be done without technical assistance	**3.53**	**0.56**	Highly Acceptable
Functions are easy to access	**3.57**	**0.54**	Highly Acceptable
Easy to find and update records	**3.56**	**0.55**	Highly Acceptable
Composite Mean	**3.56**		**Highly Acceptable**

Legend: 1.00–1.75 = Not Acceptable, 1.76–2.50 = Slightly Acceptable, 2.51–3.25 = Acceptable, 3.26–4.00 = Highly Acceptable.

Table 19 shows the respondents' attitude toward using the system. Most indicators received high ratings, reflecting that users have a positive outlook toward the platform. They reported enjoying its use, trusting its reliability, and feeling confident when operating it. The "Highly Acceptable" results indicate that users are comfortable integrating the system into their workflow.

Table 19. Attitude Toward Using the System

Indicator	Mean	Standard Deviation	Interpretation
I enjoy using the system	**3.56**	**0.54**	Highly Acceptable
The system is reliable and professional	**3.55**	**0.55**	Highly Acceptable
The system makes tasks easier	**3.54**	**0.56**	Highly Acceptable
I feel confident using the system	**3.55**	**0.55**	Highly Acceptable
I have a positive attitude toward the system	**3.57**	**0.54**	Highly Acceptable
Composite Mean	**3.55**		**Highly Acceptable**

Legend: 1.00–1.75 = Not Acceptable, 1.76–2.50 = Slightly Acceptable, 2.51–3.25 = Acceptable, 3.26–4.00 = Highly Acceptable.

Table 20 shows the respondents' intention to continue using the system. Ratings across all indicators were high, suggesting strong willingness among users to adopt the platform long-term. Respondents expressed that they plan to continue using it, prefer it over manual processes, and would recommend it to others. This reflects a strong level of acceptance and readiness for continued system adoption.

Table 20. Behavioral Intention to Use the System

Indicator	Mean	Standard Deviation	Interpretation
I plan to continue using the system	**3.59**	**0.53**	Highly Acceptable
I would recommend the system to others	**3.57**	**0.54**	Highly Acceptable
I prefer this over manual record-keeping	**3.56**	**0.55**	Highly Acceptable
I am likely to depend on this system	**3.56**	**0.55**	Highly Acceptable
I will encourage others to use it	**3.58**	**0.54**	Highly Acceptable
Composite Mean	**3.57**		**Highly Acceptable**

Legend: 1.00–1.75 = Not Acceptable, 1.76–2.50 = Slightly Acceptable, 2.51–3.25 = Acceptable, 3.26–4.00 = Highly Acceptable.

Table 21 shows the overall summary of the Technology Acceptance Model (TAM) evaluation results. The system received consistently high ratings across all four areas and the respondents indicated that the system is useful in supporting their tasks, easy to operate, reliable, and something they are willing to continue using. These results reflect a strong level of acceptance and show that users recognize the system as beneficial and suitable for ongoing use in health center operations.

Table 21. Summary of TAM Evaluation Results

TAM Construct	Mean	Standard Deviation	Interpretation
Perceived Usefulness	**3.55**	**0.56**	Highly Acceptable
Perceived Ease of Use	**3.56**	**0.55**	Highly Acceptable
Attitude Toward Using	**3.55**	**0.55**	Highly Acceptable
Behavioral Intention to Use	3.57	0.54	Highly Acceptable
Composite Mean	**3.56**		**Highly Acceptable**

Legend: 1.00–1.75 = Not Acceptable, 1.76–2.50 = Slightly Acceptable, 2.51–3.25 = Acceptable, 3.26–4.00 = Highly Acceptable.

CHAPTER V

SUMMARY, CONCLUSION AND RECOMMENDATION

This chapter presents a summary of the study, the insights gained throughout its conduct, and the recommendations that may guide future system enhancements and further development. The content reflects the overall findings and the project's contribution to improving patient record management, disease tracking, and forecasting within rural health settings.

Summary

This study developed a Patient Record System with Disease Tracking and ARIMA-Based Forecasting for Communicable Diseases, a web-based platform designed to improve patient record management, strengthen disease monitoring, and provide short-term disease trend forecasting in rural health units and community-based health centers. The project was initiated in response to the persistent challenges in manual record-keeping, delayed disease reporting, and the absence of analytical tools that could support proactive health planning. Rural Health Units in the Philippines continue to rely on paper-based systems that are prone to errors, difficult to update, and inefficient during outbreaks. These problems highlighted the need for a centralized digital platform capable of improving accuracy, accessibility, and timeliness of health data.

The system developed in this study integrates several core modules to address these concerns. The Patient Module provides structured recording of patient information, allowing health workers to store complete personal, demographic, and clinical data in one accessible location. This digital approach

reduces the possibility of misplaced files and ensures that medical staff can retrieve patient information when needed.

The Disease Tracking Module plays a central role in monitoring the top five communicable diseases commonly reported in local health units, specifically measles, chickenpox, hepatitis, tuberculosis, and dengue. Through map-based visualization and case recording, the system assists health workers in identifying hotspots, observing disease movement, and making timely decisions for intervention.

Another key component of the system is the ARIMA-Based Forecasting Module, which processes historical disease data to generate short-term projections. These forecasts assist health authorities in anticipating fluctuations in disease cases, preparing necessary resources, and implementing early interventions when needed. By integrating time-series forecasting, the platform supports evidence-based decision-making and contributes to strengthening local public health preparedness.

The system aligns with the Sustainable Development Goals (SDGs), particularly SDG 3 (Good Health and Well-Being) through improved disease surveillance and timely health information; SDG 9 (Industry, Innovation, and Infrastructure) by promoting the development of digital health technologies; and SDG 11 (Sustainable Cities and Communities) by supporting safer, more resilient communities through efficient disease monitoring and health data management.

The scope of the system focuses on digitizing patient records, tracking the top five communicable diseases, and generating short-term forecasts

based on available RHU data. It is designed to store up to five years' worth of information and visualize cases through a map-based interface. However, the system is limited by its reliance on internet connectivity for full functionality and its current coverage of only five diseases. These limitations can be addressed in future enhancements, such as integrating offline features, expanding disease coverage, and optimizing system performance for broader implementation.

Conclusion

The development of the Patient Record System with Disease Tracking and ARIMA-Based Forecasting has shown how digital solutions can significantly improve the way rural health units manage patient information and monitor communicable diseases. Guided by the study's objectives, the system was created to assist in organizing patient records, support basic case monitoring, and offer simple insights through short-term forecasting.

One of the major accomplishments of the project is the creation of a patient record module that helps ease some of the challenges associated with paper-based documentation. Through a centralized digital platform, healthcare workers are now able to record, update, and retrieve patient information more efficiently. This reduces the risks of misplaced documents and ensures that essential health data is accessible during consultations and emergency situations.

Another important component is the disease tracking feature. With map-based visualization and systematic case logging, health staff can observe patterns in disease distribution, identify affected locations, and monitor the top five communicable diseases more clearly. These capabilities support better

surveillance practices and provide health workers with timely information for community-level decision-making.

The integration of ARIMA-based forecasting further strengthens the system's contribution to public health. By analyzing historical case data, the forecasting module offers short-term predictions that can help health personnel anticipate fluctuations in disease incidence. These insights assist decision-makers in preparing necessary resources, planning field activities, and reinforcing preventive measures, which collectively enhance the RHU's readiness and response.

Another point to consider, the system features were assessed using ISO 25010 software quality standards and user feedback. The results indicated that the system meets essential criteria for functionality, usability, reliability, and performance. Healthcare staff noted that the interface is manageable and the system's features are practical for their daily tasks, suggesting that the platform is well-suited for real-world RHU operations.

To conclude, the Patient Record System with Disease Tracking and ARIMA-Based Forecasting fulfills its intended goals while contributing to broader public health initiatives. It supports SDG 3 (Good Health and Well-Being) through improved disease surveillance, SDG 9 (Industry, Innovation, and Infrastructure) by promoting the use of digital technologies in local health settings, and SDG 11 (Sustainable Cities and Communities) by helping communities become more prepared and resilient. With continued refinement, proper implementation, and institutional support, the system has the potential

to serve as a dependable model for digital health solutions in rural areas and strengthen the management of communicable diseases.

Recommendations

Based on the findings and outcomes of the study, the following recommendations are proposed to further enhance the Patient Record System with Disease Tracking and ARIMA-Based Forecasting and to expand its potential use in local health settings.

Extend the system's implementation beyond Rural Health Units. Since many barangay health centers also rely on manual record-keeping and face similar challenges in disease monitoring, the system may be adapted for use at the barangay level. Expanding the platform to smaller community-based facilities can strengthen local surveillance and create a more connected health information network.

Adopt a centralized database for real-time data sharing and consolidation. Integrating a unified database accessible to authorized health facilities would allow faster reporting, smoother information flow, and more coordinated health responses. A centralized setup can also help reduce duplicate entries and make it easier to track patient records across different locations when necessary.

Explore the use of more advanced forecasting models for improved accuracy. While the current ARIMA model provides useful short-term projections, future studies may consider incorporating advanced approaches such as SARIMA, SEIR, or hybrid time-series models to enhance predictive

performance. These models can offer more detailed insights, especially for diseases influenced by seasonal patterns or varying transmission dynamics.

Expand the system to include more communicable diseases and additional barangays or locations. To strengthen the system's coverage further, future developers may consider extending the monitoring and forecasting features to a wider range of communicable diseases beyond the initial five included in this study. And also, increasing the number of barangays or locations within the system's scope would provide a more comprehensive view of disease trends across larger communities. This enhancement can help local health offices generate broader and more detailed analyses, enabling more accurate planning and targeted intervention in various areas.

Utilize more accurate and complete datasets using actual raw case data. Although the system performed effectively using the aggregated data provided, future improvements may benefit from the use of more precise and detailed datasets. Access to actual raw case data while still adhering to the Data Privacy Act can significantly improve the accuracy of disease modeling, trend analysis, and forecasting results. With a richer dataset, the system can better capture real patterns within the community, helping produce more reliable insights and supporting improved decision-making for health authorities.

These recommendations aim to support future enhancements and guide the continuous development of the system, ensuring that it remains relevant, adaptable, and beneficial to the needs of local health units and communities.

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APPENDIX A.

Letter of Approval

CERTIFICATE OF APPROVAL

This is to certify that after having satisfactorily completed all the requirements of the capstone project entitled:

“PATIENT RECORD SYSTEM WITH DISEASE TRACKING AND ARIMA-BASED FORECASTING FOR TOP 5 DISEASES IN LOS BAÑOS, LAGUNA”

ARIEL LONGA, RYAN ERICK FAUSTINO, and MARK BRIAN NAVARRO have been granted permission to proceed with their proposal presentation.

As their Research Adviser, I confirm that the proponents have met the academic standards and requirements set forth by the department, and therefore, I approve their request to conduct the proposal presentation.

SHERWIN B. SAPIN
Capstone Project Adviser

APPENDIX B.

Source Code

ARIMA Model

```
import pandas as pd
import numpy as np
from statsmodels.tsa.arima.model import ARIMA
from sklearn.metrics import mean_squared_error, mean_absolute_percentage_error
import matplotlib.pyplot as plt
import os
import json
import sys
import pickle
import time
from datetime import datetime, timedelta

script_dir = os.path.dirname(os.path.abspath(__file__))
os.chdir(script_dir)

def preprocess_data(data):
    data['year'] = data['year'].astype(int)
    data['month'] = data['month'].astype(int)
    data['total_cases'] = pd.to_numeric(data['total_cases'], errors='coerce')
    data["date"] = pd.to_datetime(data["year"].astype(str) + "-" + data["month"].astype(str) + "-01")
    data['total_cases'] = data['total_cases'].fillna(method='ffill').fillna(method='bfill').fillna(0)
    data = data.dropna()
    data = data.sort_values('date')
    return data

def load_cached_model(disease, forecast_period):
    cache_file = f"cache/{disease}_{forecast_period}.pkl"
    if os.path.exists(cache_file) and (time.time() - os.path.getmtime(cache_file)) < 3600:
        try:
            with open(cache_file, 'rb') as f:
                return pickle.load(f)
        except:
            return None
    return None

def save_cached_model(model, disease, forecast_period):
    cache_file = f"cache/{disease}_{forecast_period}.pkl"
    os.makedirs(os.path.dirname(cache_file), exist_ok=True)
    with open(cache_file, 'wb') as f:
        pickle.dump(model, f)

def calculate_accuracy_metrics(actual, predicted):
    mask = ~(np.isnan(actual) | np.isnan(predicted))
    actual_clean = actual[mask]
    predicted_clean = predicted[mask]

    if len(actual_clean) == 0:
        return {'rmse': 0, 'mape': 0}

    rmse = np.sqrt(mean_squared_error(actual_clean, predicted_clean))
    mape = mean_absolute_percentage_error(actual_clean, predicted_clean) * 100
    return {'rmse': float(rmse), 'mape': float(mape)}

def main():
    try:
        import warnings
        warnings.filterwarnings('ignore')

        if len(sys.argv) < 2:
            print(json.dumps({"success": False, "error": "JSON file path required"}))
            return

        json_file = sys.argv[1]
        forecast_period = int(sys.argv[2]) if len(sys.argv) > 2 else 3

        with open(json_file, 'r') as f:
            data_list = json.load(f)

        data = pd.DataFrame(data_list)
```

```
required_columns = {"year", "month", "disease_name", "total_cases"}
if not required_columns.issubset(data.columns):
    raise ValueError("Dataset must contain year, month, disease_name, total_cases columns.")

data = preprocess_data(data)

current_date = datetime.now()
thirty_days_ago = current_date - timedelta(days=30)

recent_data = data[data['date'] >= thirty_days_ago]

current_cases_30d = {}
if not recent_data.empty:
    for disease, group in recent_data.groupby('disease_name'):
        current_cases_30d[disease] = int(group['total_cases'].sum())

forecast_results = []

for disease, group in data.groupby("disease_name"):
    group = group.sort_values("date")
    group.set_index("date", inplace=True)
    ts = group["total_cases"]

    if len(ts) < 5:
        forecast_results.append({
            "disease_name": disease,
            "forecast_month": "Error",
            "forecast_cases": 0,
            "error": "Not enough data points for ARIMA forecast"
        })
        continue

    try:
        cached_model = load_cached_model(disease, forecast_period)

        if cached_model is not None:
            model_fit = cached_model
        else:
            model = ARIMA(ts, order=(1, 1, 1))
            model_fit = model.fit()

            save_cached_model(model_fit, disease, forecast_period)

        forecast = model_fit.forecast(steps=forecast_period)

        if len(ts) >= 10:
            split_point = int(len(ts) * 0.8)
            train_data = ts[:split_point]
            test_data = ts[split_point:]

            temp_model = ARIMA(train_data, order=(1, 1, 1))
            temp_fit = temp_model.fit()
            validation_forecast = temp_fit.forecast(steps=len(test_data))

            accuracy_metrics = calculate_accuracy_metrics(test_data.values, validation_forecast)
        else:
            accuracy_metrics = {'rmse': 0, 'mape': 0}

        current_date = datetime.now()
        for i, val in enumerate(forecast):
            next_month = pd.Timestamp(current_date) + pd.DateOffset(months=i + 1)
            forecast_results.append({
                "disease_name": disease,
                "forecast_month": next_month.strftime("%Y-%m"),
                "forecast_cases": int(round(float(val))) if not pd.isna(val) else int(max(1, float(ts.mean()))),
                "accuracy_rmse": accuracy_metrics['rmse'],
                "accuracy_mape": accuracy_metrics['mape']
            })
```

```
            plt.figure(figsize=(8, 4))
            plt.plot(ts, label="Actual Cases", marker='o')
            forecast_dates = pd.date_range(pd.Timestamp(current_date), periods=forecast_period+1, freq="M")[1:]
            plt.plot(forecast_dates, forecast, label="Forecast", linestyle="--", marker='s')
            plt.title(f"Disease Forecast: {disease}")
            plt.xlabel("Date")
            plt.ylabel("Cases")
            plt.legend()
            plt.grid(True)
            plt.tight_layout()
            plt.savefig(f"forecast_{disease.lower().replace(' ', '_')}.png", dpi=300, bbox_inches='tight')
            plt.close()

        except Exception as e:
            forecast_results.append({
                "disease_name": disease,
                "forecast_month": "Error",
                "forecast_cases": 0,
                "error": str(e)
            })

    result_data = {
        'success': True,
        'forecast_results': forecast_results,
        'summary': {
            'total_diseases': int(len(set(r['disease_name'] for r in forecast_results if 'error' not in r))),
            'total_forecast_months': int(len(forecast_results)),
            'historical_records': int(len(data)),
            'current_cases_30d': current_cases_30d,
            'generated_at': datetime.now().strftime('%Y-%m-%d %H:%M:%S')
        }
    }

    print(json.dumps(result_data))
    return forecast_results

  except Exception as e:
    error_data = {
        'success': False,
        'error': str(e)
    }
    print(json.dumps(error_data))
    return None

if __name__ == "__main__":
  main()
```

Backend

```
<?php
error_reporting(0);
ini_set('display_errors', 0);

require 'cors.php';
require 'config.php';

$data = json_decode(file_get_contents("php:

if (
  !isset($data['first_name']) || empty($data['first_name']) ||
  !isset($data['surname']) || empty($data['surname']) ||
  !isset($data['date_of_birth']) || empty($data['date_of_birth']) ||
  !isset($data['sex']) || empty($data['sex']) ||
  !isset($data['address']) || empty($data['address'])
) {
  http_response_code(400);
  echo json_encode(['error' => 'Missing required fields: First Name, Surname, Date of Birth, Gender, and Address
      are required.']);
  exit;
}
```

```
$first_name = $data['first_name'];
$middle_name = $data['middle_name'] ?? '';
$surname = $data['surname'];
$suffix = $data['suffix'] ?? '';

$full_name = trim($first_name . ' ' . $middle_name . ' ' . $surname . ' ' . $suffix);

$date_of_birth = $data['date_of_birth'];
$sex = $data['sex'];
$address = $data['address'];

$philhealth_id = $data['philhealth_id'] ?? '';
$priority = $data['priority'] ?? 'medium';

$birthDate = new DateTime($date_of_birth);
$today = new DateTime();
$age = $today->diff($birthDate)->y;

$stmt = $conn->prepare("INSERT INTO patients (full_name, age, sex, date_of_birth, address) VALUES (?, ?, ?, ?,
        ?)");
$stmt->bind_param("sisss", $full_name, $age, $sex, $date_of_birth, $address);

if ($stmt->execute()) {
    $newId = $conn->insert_id;

    $medical_stmt = $conn->prepare("INSERT INTO medical_records (patient_id, surname, first_name, middle_name,
        suffix, philhealth_id, priority, created_at, updated_at) VALUES (?, ?, ?, ?, ?, ?, ?, NOW(), NOW())");
    $medical_stmt->bind_param("isssssss", $newId, $surname, $first_name, $middle_name, $suffix, $philhealth_id,
        $priority);

    $medical_stmt->execute();

    echo json_encode([
        'success' => true,
        'data' => [
            'id' => $newId,
            'full_name' => $full_name,
            'sex' => $sex,
            'age' => $age,
            'address' => $address
        ]
    ]);
} else {
    http_response_code(500);
    echo json_encode(['success' => false, 'error' => 'Database error: ' . $stmt->error]);
}
?>



<?php
require 'cors.php';
require 'config.php';

$data = json_decode(file_get_contents("php:
$id = (int)($data['id'] ?? 0);

if (!$id) {
    http_response_code(400);
    echo json_encode(['error' => 'Missing patient ID']);
    exit;
}

$sql = "DELETE FROM patients WHERE id = $id";

if (mysqli_query($conn, $sql)) {
    echo json_encode(['success' => true]);
} else {
    http_response_code(500);
    echo json_encode(['error' => 'Database error: ' . mysqli_error($conn)]);
}
```

```
<?php
require 'cors.php';
require 'config.php';

$page = isset($_GET['page']) ? (int)$_GET['page'] : 1;
$limit = isset($_GET['limit']) ? (int)$_GET['limit'] : 25;
$search = isset($_GET['search']) ? trim($_GET['search']) : '';
$disease = isset($_GET['disease']) ? trim($_GET['disease']) : '';
$sortBy = isset($_GET['sortBy']) ? $_GET['sortBy'] : 'id';
$sortOrder = isset($_GET['sortOrder']) ? $_GET['sortOrder'] : 'asc';

$page = max(1, $page);
$limit = max(1, min(100, $limit));
$offset = ($page - 1) * $limit;

$allowedSortFields = ['id', 'full_name', 'age', 'sex', 'address', 'created_at', 'last_visit_date'];
if (!in_array($sortBy, $allowedSortFields)) {
    $sortBy = 'id';
}

$sortOrder = strtoupper($sortOrder) === 'DESC' ? 'DESC' : 'ASC';

$whereConditions = [];
$params = [];
$paramTypes = '';

if (!empty($search)) {
    $whereConditions[] = "(p.full_name LIKE ? OR p.address LIKE ? OR p.sex LIKE ?)";
    $searchParam = "%$search%";
    $params[] = $searchParam;
    $params[] = $searchParam;
    $params[] = $searchParam;
    $paramTypes .= 'sss';
}

if (!empty($disease) && $disease !== 'all') {
    if ($disease === 'healthy') {
        $whereConditions[] = "(mr.diagnosis IS NULL OR mr.diagnosis = '' OR mr.diagnosis = 'Healthy')";
    } else {
        $whereConditions[] = "mr.diagnosis = ?";
        $params[] = $disease;
        $paramTypes .= 's';
    }
}

$searchCondition = !empty($whereConditions) ? "WHERE " . implode(' AND ', $whereConditions) : "";


$countSql = "
    SELECT COUNT(*) as total
    FROM patients p
    LEFT JOIN (
        SELECT
            patient_id,
            diagnosis,
            ROW_NUMBER() OVER (PARTITION BY patient_id ORDER BY updated_at DESC) as rn
        FROM medical_records
    ) mr ON p.id = mr.patient_id AND mr.rn = 1
    $searchCondition
";

$countStmt = $conn->prepare($countSql);
if (!$countStmt) {
    http_response_code(500);
    echo json_encode(['error' => 'Count prepare failed: ' . $conn->error]);
    exit;
}

if (!empty($params)) {
    $countStmt->bind_param($paramTypes, ...$params);
}

if (!$countStmt->execute()) {
    http_response_code(500);
    echo json_encode(['error' => 'Count execute failed: ' . $countStmt->error]);
```

```
    exit;
}

$countResult = $countStmt->get_result();
$totalRecords = $countResult->fetch_assoc()['total'];
$totalPages = ceil($totalRecords / $limit);

$sql = "
    SELECT
        p.*,
        mr.diagnosis,
        COALESCE(
            mr.date_of_consultation,
            mr.date_of_consultation_medical,
            mr.updated_at,
            p.created_at
        ) as last_visit_date
    FROM patients p
    LEFT JOIN (
        SELECT
            patient_id,
            diagnosis,
            date_of_consultation,
            date_of_consultation_medical,
            updated_at,
            ROW_NUMBER() OVER (PARTITION BY patient_id ORDER BY updated_at DESC) as rn
        FROM medical_records
    ) mr ON p.id = mr.patient_id AND mr.rn = 1
    $searchCondition
    ORDER BY p.$sortBy $sortOrder
    LIMIT ? OFFSET ?
";

$stmt = $conn->prepare($sql);
if (!$stmt) {
    http_response_code(500);
    echo json_encode(['error' => 'Prepare failed: ' . $conn->error]);
    exit;
}

$allParams = $params;
$allParams[] = $limit;
$allParams[] = $offset;
$allParamTypes = $paramTypes . 'ii';

if (!empty($allParams)) {
    $stmt->bind_param($allParamTypes, ...$allParams);
}

if (!$stmt->execute()) {
    http_response_code(500);
    echo json_encode(['error' => 'Execute failed: ' . $stmt->error]);
    exit;
}

$result = $stmt->get_result();
$patients = [];
while ($row = $result->fetch_assoc()) {
    $patients[] = $row;
}

echo json_encode([
    'success' => true,
    'data' => $patients,
    'pagination' => [
        'currentPage' => $page,
        'totalPages' => $totalPages,
        'totalRecords' => $totalRecords,
        'limit' => $limit,
        'hasNext' => $page < $totalPages,
        'hasPrev' => $page > 1
    ]
]);
?>
```

```
<?php
require 'cors.php';
require 'config.php';

header("Content-Type: application/json");

try {
    $sql = "SELECT id, disease, forecast_type, forecast_period, population, forecast_results,
                indicators, area_data, current_data, generated_at
            FROM forecasts
            ORDER BY generated_at DESC
            LIMIT 10";

    $result = $conn->query($sql);
    $forecasts = [];

    while ($row = $result->fetch_assoc()) {
        $forecast_results = json_decode($row['forecast_results'], true);

        $predictions = [];
        if ($forecast_results && is_array($forecast_results)) {
            foreach ($forecast_results as $forecast) {
                if (isset($forecast['forecast_month']) && isset($forecast['forecast_cases'])) {
                    $predictions[] = [
                        'month' => $forecast['forecast_month'],
                        'cases' => $forecast['forecast_cases']
                    ];
                }
            }
        }

        $forecasts[] = [
            'id' => $row['id'],
            'disease' => $row['disease'],
            'forecast_type' => $row['forecast_type'],
            'forecast_period' => $row['forecast_period'],
            'population' => $row['population'],
            'forecast_results' => $forecast_results,
            'predictions' => $predictions,
            'indicators' => json_decode($row['indicators'], true),
            'area_data' => json_decode($row['area_data'], true),
            'current_data' => json_decode($row['current_data'], true),
            'generated_at' => $row['generated_at']
        ];
    }

    echo json_encode([
        'success' => true,
        'data' => $forecasts,
        'count' => count($forecasts)
    ]);

} catch (Exception $e) {
    http_response_code(500);
    echo json_encode([
        'success' => false,
        'error' => $e->getMessage()
    ]);
}
?>



<?php
error_reporting(0);
ini_set('display_errors', 0);

require 'cors.php';
require 'config.php';

$data = json_decode(file_get_contents("php:

if (
    !isset($data['id']) ||
```

```
    !isset($data['full_name']) || empty($data['full_name']) ||
    !isset($data['date_of_birth']) || empty($data['date_of_birth']) ||
    !isset($data['sex']) || empty($data['sex']) ||
    !isset($data['address']) || empty($data['address'])
) {
    http_response_code(400);
    echo json_encode(['error' => 'Missing required fields: Full Name, Date of Birth, Gender, and Address are required.']);
    exit;
}

$id = (int)$data['id'];
$full_name = $data['full_name'];
$date_of_birth = $data['date_of_birth'];
$sex = $data['sex'];
$address = $data['address'];

$birthDate = new DateTime($date_of_birth);
$today = new DateTime();
$age = $today->diff($birthDate)->y;

$stmt = $conn->prepare("UPDATE patients SET full_name = ?, date_of_birth = ?, sex = ?, age = ?, address = ?
        WHERE id = ?");
$stmt->bind_param("sssisi", $full_name, $date_of_birth, $sex, $age, $address, $id);

if ($stmt->execute()) {
    echo json_encode([
        'success' => true,
        'data' => [
            'id' => $id,
            'full_name' => $full_name,
            'date_of_birth' => $date_of_birth,
            'sex' => $sex,
            'age' => $age,
            'address' => $address
        ]
    ]);
} else {
    http_response_code(500);
    echo json_encode(['success' => false, 'error' => 'Database error: ' . $stmt->error]);
}
?>
```

Frontend

```
import React, { useState, useEffect, Suspense, lazy } from "react";
import { Routes, Route, Navigate, useLocation, useNavigate } from "react-router-dom";
import Sidebar from "./components/Sidebar";
import TopBar from "./components/TopBar";
import StaffTopBar from "./staff/components/StaffTopBar";
import Login from "./admin/Login";
import SessionManager from "./components/SessionManager";
import StaffSessionManager from "./staff/components/StaffSessionManager";
import { BackupProvider } from "./contexts/BackupContext";
import "./App.css";
import "./index.css";

import { preloadAllRoutes } from './utils/routePreloader';
import { preloadAllData } from './utils/dataPreloader';
import { clearCache } from './utils/cache';


const Dashboard = lazy(() => import('./admin/Dashboard'));
const Patient = lazy(() => import('./admin/Patient'));
const Records = lazy(() => import('./admin/Records'));
const Diseases = lazy(() => import('./admin/Diseases'));
const Tracker = lazy(() => import('./admin/Tracker'));
const ARIMAForecast = lazy(() => import('./admin/ARIMAForecast'));
const Reports = lazy(() => import('./admin/Reports'));
const AuditLogs = lazy(() => import('./admin/AuditLogs'));
const Settings = lazy(() => import('./admin/Settings'));
const NotificationCenter = lazy(() => import('./components/NotificationCenter'));
```

```
const RequireAdmin = lazy(() => import('./components/RequireAdmin'));
const RequireStaff = lazy(() => import('./staff/components/RequireStaff'));
const StaffLayout = lazy(() => import('./staff/layouts/StaffLayout'));
const StaffDashboard = lazy(() => import('./staff/pages/Dashboard'));
const StaffPatients = lazy(() => import('./staff/pages/Patients'));
const StaffRecords = lazy(() => import('./staff/pages/Records'));
const StaffDiseases = lazy(() => import('./staff/pages/StaffDiseases'));
const StaffTracker = lazy(() => import('./staff/pages/Tracking'));
const StaffReports = lazy(() => import('./staff/pages/Reports'));
const StaffSettings = lazy(() => import('./staff/pages/Settings'));
const StaffARIMAForecast = lazy(() => import('./staff/pages/StaffARIMAForecast'));


const PageLoader = () => (
  <div className="flex items-center justify-center min-h-screen">
    <div className="animate-spin rounded-full h-32 w-32 border-b-2 border-blue-500"></div>
  </div>
);

function App() {
  const location = useLocation();
  const navigate = useNavigate();
  const [isLoggedIn, setIsLoggedIn] = useState(() => {
    return sessionStorage.getItem("isLoggedIn") === "true";
  });
  const [sidebarCollapsed, setSidebarCollapsed] = useState(() => {
    return localStorage.getItem("sidebarCollapsed") === "true";
  });
  const [staffSidebarCollapsed, setStaffSidebarCollapsed] = useState(() => {
    return localStorage.getItem("staffSidebarCollapsed") === "true";
  });
  const [isTransitioning, setIsTransitioning] = useState(false);
  const [isLoggingOut, setIsLoggingOut] = useState(false);

  const toggleSidebar = () => {
    const newState = !sidebarCollapsed;
    setSidebarCollapsed(newState);
    localStorage.setItem("sidebarCollapsed", newState.toString());
  };

  const toggleStaffSidebar = () => {
    const newState = !staffSidebarCollapsed;
    setStaffSidebarCollapsed(newState);
    localStorage.setItem("staffSidebarCollapsed", newState.toString());
  };


  useEffect(() => {
    if (isLoggedIn) {
      const initializeApp = async () => {

        preloadAllRoutes();


        preloadAllData();
      };

      initializeApp();
    }


    const handleLogout = () => {
      setIsLoggingOut(true);
      clearCache();
    };

    window.addEventListener('logout', handleLogout);
    return () => window.removeEventListener('logout', handleLogout);
  }, [isLoggedIn]);

  const handleLogin = (user) => {
    setIsTransitioning(true);


    setTimeout(() => {
```

```
    sessionStorage.setItem("isLoggedIn", "true");
    setIsLoggedIn(true);

    try {

      localStorage.removeItem('staff_token');
      localStorage.removeItem('staff_role');
      localStorage.removeItem('admin_role');
      localStorage.removeItem('admin_token');

      if (user?.role === 'staff') {

        localStorage.setItem('staff_token', 'test-staff-token');
        localStorage.setItem('staff_role', 'staff');
        navigate('/staff/dashboard', { replace: true });
      } else {

        localStorage.setItem('admin_token', 'test-admin-token');
        localStorage.setItem('admin_role', 'admin');
        navigate('/', { replace: true });
      }
    } catch {}


    setTimeout(() => {
      setIsTransitioning(false);



      setTimeout(() => {
        window.dispatchEvent(new CustomEvent('login'));
      }, 500);
    }, 300);
  }, 100);
 };

 if (!isLoggedIn) {
   return <Login onLogin={handleLogin} />;
 }

 const isStaffRoute = location.pathname.startsWith('/staff');


 const appContent = (
   <div className="app-layout">
     {!isStaffRoute && <Sidebar collapsed={sidebarCollapsed} />}
     {!isStaffRoute && <TopBar userId={1} userName="Admin" userRole="Administrator"
onToggleSidebar={toggleSidebar} sidebarCollapsed={sidebarCollapsed} />}
     {isStaffRoute && <StaffTopBar onToggleSidebar={toggleStaffSidebar}
sidebarCollapsed={staffSidebarCollapsed} />}
     <div className={`main-content ${isStaffRoute ? (staffSidebarCollapsed ? 'sidebar-collapsed' : '') :
(sidebarCollapsed ? 'sidebar-collapsed' : '')} ${isTransitioning ? 'page-enter' : ''} ${isLoggingOut ? 'page-exit' :
''}`}>
       <Suspense fallback={<PageLoader />}>
           <Routes>
           <Route path="/" element={<RequireAdmin><Dashboard /></RequireAdmin>} />
           <Route path="/records" element={<RequireAdmin><Records /></RequireAdmin>} />
           <Route path="/patient" element={<RequireAdmin><Patient /></RequireAdmin>} />
           <Route path="/diseases" element={<RequireAdmin><Diseases /></RequireAdmin>} />
           <Route path="/tracker" element={<RequireAdmin><Tracker /></RequireAdmin>} />
           <Route path="/arima-forecast" element={<RequireAdmin><ARIMAForecast /></RequireAdmin>} />
           <Route path="/reports" element={<RequireAdmin><Reports /></RequireAdmin>} />
           <Route path="/audit-logs" element={<RequireAdmin><AuditLogs /></RequireAdmin>} />
           <Route path="/settings" element={<RequireAdmin><Settings /></RequireAdmin>} />
           <Route path="/notifications" element={<RequireAdmin><NotificationCenter /></RequireAdmin>} />
           {}
           <Route
             path="/staff/*"
             element={
               <RequireStaff>
                 <StaffLayout sidebarCollapsed={staffSidebarCollapsed} />
               </RequireStaff>
             }
           >
             <Route index element={<Navigate to="dashboard" replace />} />
```

```
              <Route path="dashboard" element={<StaffDashboard />} />
              <Route path="patients" element={<StaffPatients />} />
              <Route path="records" element={<StaffRecords />} />
              <Route path="diseases" element={<StaffDiseases />} />
              <Route path="tracking" element={<StaffTracker />} />
              <Route path="forecasts" element={<StaffARIMAForecast />} />
              <Route path="reports" element={<StaffReports />} />
              <Route path="settings" element={<StaffSettings />} />
            </Route>
            <Route path="*" element={<Navigate to="/" />} />
          </Routes>
        </Suspense>
      </div>
    </div>
  );

  return (
    <BackupProvider>
      {isStaffRoute ? (
        <StaffSessionManager>{appContent}</StaffSessionManager>
      ) : (
        <SessionManager>{appContent}</SessionManager>
      )}
    </BackupProvider>
  );
}

export default App;
```

APPENDIX C.

Evaluation Tool

ISO SURVEY QUESTIONNAIRE FOR IT SPECIALIST

Good day! We are students from Laguna State Polytechnic University Los Baños Campus, College of Computer Studies, conducting a survey for our Capstone Project titled "Patient Record System with Patient Tracking and ARIMA-Based Forecasting for Communicable Diseases." Please evaluate the system based on the given criteria following the ISO 25010 standards. Your responses will remain confidential and used only for academic purposes. Thank you for your time and participation.

The Researchers: Ariel C. Longa, Mark Brian Navarro, Ryan Erick Faustino

5 – Excellent 4 – Good 3 – Fair 2 – Needs Improvement 1 – Poor

RUBRIC FOR 5-Point Likert Scale to Evaluate System Quality Based on ISO 25010

Characteristics	Excellent (5)	Good (4)	Fair (3)	Needs Improvement (2)	Poor (1)
Functional Suitability	The system meets all required functions and performs both patient record management and forecasting tasks accurately and consistently.	Most functions work properly with only small issues that do not affect key operations..	The main functions are present but some features or forecasting results feel limited in accuracy or scope	Only a few functions work correctly and there are noticeable problems with accuracy or completeness.	The system fails to meet its main purpose and has major problems in both data management and forecasting.
Performance Efficiency	The system responds quickly and performs efficiently even with large data or multiple users.	It performs well with only minimal delays during heavy usage.	Performance is acceptable but there are times when it slows down or lags.	The system often feels slow or takes time to load certain functions.	Performance is poor and the system runs slowly or becomes unresponsive most of the time.
Compatibility	The system works well across all browsers, devices, and platforms with full functionality.	It is mostly compatible with only small issues that do not affect key features.	The system works on most platforms but shows limited support on others.	Compatibility is limited and some features fail to work in specific environments.	The system does not work properly on most devices or platforms.
Interaction Capability	The system is very easy to use and navigate. It provides a clear and friendly interface that supports user efficiency.	It is generally easy to use with only minor usability issues.	The system can be used with some adjustments or learning time.	It can be confusing to use without guidance or prior experience.	The system is difficult to use and has a poor interface design.
Reliability	The system runs smoothly without crashes or major errors	The system is generally stable with rare minor issues that do	There are occasional stability issues or short downtimes	The system experiences frequent minor crashes or noticeable	The system is unstable and often crashes or fails to load properly.

	even under heavy use.	not affect overall performance.	that slightly affect usage.	errors during normal use.	
Security	The system provides strong data protection and ensures that patient information is safe and private.	The system is secure overall, though small improvements could make it stronger.	The system has basic security features like logins and access controls but lacks advanced protection.	Some parts of the system are not secure and may expose sensitive information.	The system has weak or missing security features, making it unsafe for data storage.
Maintainability	The system is well-organized and easy to maintain, update, and debug.	It is mostly maintainable with only a few parts that require extra effort to update.	Maintenance is possible but some components are complex and hard to modify.	The system is difficult to maintain and fixing issues takes time.	The system is very hard to update or fix without major changes to its code.
Flexibility	The system can easily be installed or used in other locations or institutions without problems.	It can be transferred to another environment with only small adjustments.	It works in different setups but may need some configuration changes.	It requires significant setup or adjustments before it can run in another environment.	The system cannot be transferred or used in another environment without being reconfigured completely.
Safety	The system ensures safety by preventing data loss, errors in forecasting, and misuse of patient records.	The system is safe overall but minor issues could affect data handling accuracy.	Basic safety protocols are in place but lack advanced error prevention mechanisms.	The system sometimes risks data inaccuracies or forecasting errors.	The system poses a high risk of data loss or misuse due to missing safeguards.

_____ _____

Signature of Respondent **Date**

TECHNOLOGY ACCEPTANCE MODEL RUBRIC

We are students from Laguna State Polytechnic University – Los Baños Campus, College of Computer Studies, conducting a study for our capstone project titled "Patient Record System with Patient Tracking and ARIMA-Based Forecasting for Communicable Diseases." Please rate each statement based on your experience using the system. All information will be kept confidential and used only for academic purposes in compliance with the Data Privacy Act of 2012 (RA 10173). Thank you for your cooperation.

The Proponents:

Ariel Longa, Ryan Erick Faustino, Mark Brian Navarro

Rating Scale

4 – Highly Acceptable | 3 – Moderately Acceptable | 2 – Slightly Acceptable | 1 – Not Acceptable

Perceived Usefulness	**Highly Acceptable (4)**	**Moderately Acceptable (3)**	**Slightly Acceptable (2)**	**Not Acceptable (1)**
1. The system helps me complete my work efficiently.	Greatly increases productivity and accuracy in performing tasks.	Improves work speed and accuracy with few limitations.	Helps but only slightly affects productivity.	Does not help improve task efficiency.
2. The system improves management of patient data and records.	Strongly enhances organization and accessibility of records.	Improves data organization with minor issues.	Helps but data management still needs improvement.	Does not improve data management.
3. The forecasting feature helps in making better health-related decisions.	Provides accurate forecasts that improve decision-making.	Forecasting helps decision-making with few inconsistencies.	Forecasting gives limited help in planning or decisions.	Forecasting is unreliable or not useful.
4. The system is useful for monitoring and controlling disease cases.	Significantly helps in tracking and managing disease data.	Assists monitoring with minor data delays or errors.	Helps occasionally but lacks full reliability.	Does not aid in disease monitoring.
5. The system contributes to faster response and decision-making in health management.	Strongly supports quick and informed actions in health-related tasks.	Moderately improves decision-making speed.	Slightly influences response time.	Does not help improve response or decision speed.
Perceived Ease of Use	**Highly Acceptable (4)**	**Moderately Acceptable (3)**	**Slightly Acceptable (2)**	**Not Acceptable (1)**
1. The system is easy to learn and operate.	Very easy to learn and use without assistance.	Easy to learn with minimal guidance.	Some functions are difficult to learn.	Hard to learn and operate.
2. The interface is simple, organized, and user-friendly.	Layout is clean, intuitive, and efficient.	Interface is clear with minor usability issues.	Interface is cluttered or confusing at times.	Interface is poorly designed and hard to use.
3. Tasks can be done without much technical support.	Fully independent in performing tasks.	Occasionally needs technical help.	Needs frequent assistance.	Cannot operate without help.
4. Functions are easy to access and understand.	Functions are straightforward and logically arranged.	Most features are easy to access.	Some features are confusing or hidden.	Functions are difficult to locate or understand.

5. It is easy to find, view, and update patient records.	Record access and updating are fast and reliable.	Accessible with minor slowdowns.	Sometimes it is hard to search or edit records.	Very difficult to find or update records.
Attitude Toward Using	**Highly Acceptable (4)**	**Moderately Acceptable (3)**	**Slightly Acceptable (2)**	**Not Acceptable (1)**
1. I enjoy using the system in performing my work.	Highly enjoyable and satisfying to use.	Generally enjoyable with few frustrations.	Neutral feeling toward the system.	Unpleasant to use.
2. I find the system reliable and professional.	Consistently reliable and professional-looking.	Reliable most of the time.	Occasionally unreliable or unpolished.	Unreliable and unprofessional.
3. The system makes my tasks easier to accomplish.	Greatly simplifies work processes.	Simplifies some tasks effectively.	Helps a little but not much.	Does not make tasks easier.
4. I feel confident using the system daily.	Very confident and comfortable using it.	Confident most of the time.	Sometimes unsure using it.	Not confident using it.
5. I have a positive attitude toward using the system.	Feels motivated and positive using the system.	Generally positive with few concerns.	Neutral or indifferent.	Negative attitude toward system use.
Behavioral Intention to Use	**Highly Acceptable (4)**	**Moderately Acceptable (3)**	**Slightly Acceptable (2)**	**Not Acceptable (1)**
1. I plan to continue using the system in the future.	Strongly intends to continue using it regularly.	Likely to continue using it.	Unsure about continued use.	Does not intend to use it again.
2. I would recommend this system to others.	Highly willing to recommend to others.	Would recommend with minor reservations.	Rarely recommends it.	Would not recommend it.
3. I prefer using this system instead of manual record-keeping.	Strong preference for this system over manual methods.	Prefers system but sometimes uses manual records.	Occasionally prefers manual methods.	Prefers manual record-keeping.
4. I am likely to depend on this system for my work.	Strongly relies on the system for daily tasks.	Often depends on it.	Sometimes uses it as backup.	Does not depend on it at all.
5. I will encourage others to use this system as well.	Actively promotes system use to others.	Occasionally encourages others.	Rarely encourages use.	Does not encourage others to use it.

_____ _____

Signature of Respondent **Date**

APPENDIX D.

ISO and TAM Tally

ISO 25010 Item-Level Tally of IT Expert Responses

ISO 25010 Characteristic / Item	IT Expert 1	IT Expert 2	IT Expert 3
Functional Suitability			
Functional Completeness	5	5	5
Functional Correctness	5	5	5
Functional Appropriateness	5	5	5
Performance Efficiency			
Time Behaviour	5	5	5
Resource Utilization	5	5	5
Capacity	5	5	4
Compatibility			
Co-existence	5	5	4
Interoperability	5	5	5
Interaction Capability (Usability)			
Appropriateness Recognizability	5	5	5
Learnability	5	5	5
Operability	5	5	5
User Error Protection	5	5	5
User Engagement	5	5	5
Inclusivity	5	5	5
User Assistance	5	5	5
Self-descriptiveness	5	4	5
Reliability			
Faultlessness	5	5	5
Availability	5	5	5
Fault Tolerance	5	4	5
Recoverability	5	5	5
Security			

Confidentiality	5	5	5
Integrity	5	5	5
Non-repudiation	5	5	5
Accountability	5	5	5
Authenticity	5	5	5
Resistance	5	5	5
Maintainability			
Modularity	5	5	5
Reusability	5	5	5
Analyzability	5	5	4
Modifiability	4	5	5
Testability	5	5	4
Flexibility			
Adaptability	5	5	5
Scalability	5	5	5
Replaceability	5	5	5
Safety			
Operational Constraint	5	5	5
Risk Identification	5	5	5
Fail Safe	5	5	5
Hazard Warning	5	5	5
Safe Integration	5	5	5

TAM Item-Level Tally of User Responses

	Perceived Usefulness			
Item	**4 - Strongly Agree**	**3 - Agree**	**2 - Disagree**	**1 - Strongly Disagree**
Helps me complete work efficiently	59	37	4	0
Improves management of patient data	58	38	4	0

Forecasting helps decision-making	57	38	5	0
Useful for monitoring disease cases	62	34	4	0
Contributes to faster response and decisions	60	35	5	0

Perceived Ease of Use

Item	4 - Strongly Agree	3 - Agree	2 - Disagree	1 - Strongly Disagree
System is easy to learn	59	37	4	0
Interface is simple and user-friendly	62	34	4	0
Tasks without technical assistance	57	39	4	0
Functions are easy to access	61	35	4	0
Easy to find and update records	60	36	4	0

Attitude Toward Using

Item	4 - Strongly Agree	3 - Agree	2 - Disagree	1 - Strongly Disagree
I enjoy using the system	60	36	4	0
System is reliable and professional	59	37	4	0
System makes tasks easier	59	36	5	0
I feel confident using the system	59	37	4	0
I have a positive attitude	61	35	4	0

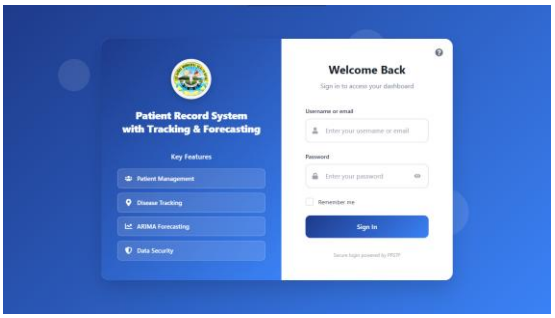
Behavioral Intention to Use

Item	4 - Strongly Agree	3 - Agree	2 - Disagree	1 - Strongly Disagree
I plan to continue using the system	63	33	4	0
I would recommend the system	61	35	4	0
Prefer over manual record-keeping	60	36	4	0
Likely to depend on the system	60	36	4	0
Encourage others to use the system	62	34	4	0

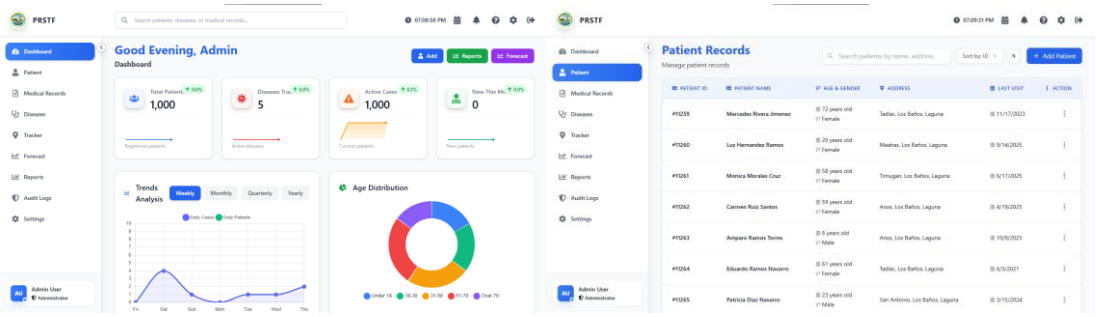
APPENDIX E.

Sample Output

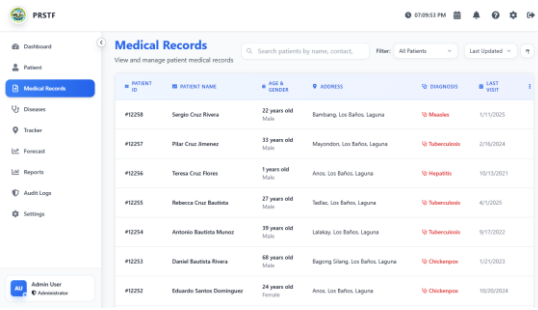
Login

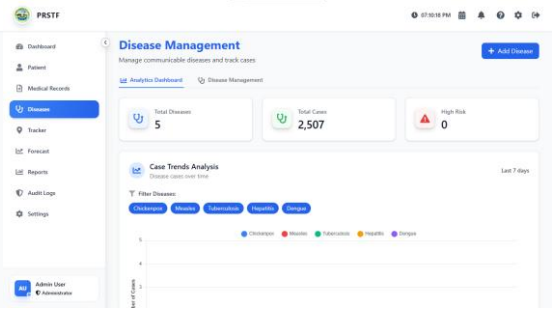

Patient Record System
with Tracking & Forecasting
Key Features
Patient Management
Disease Tracking
ARIMA Forecasting
Data Security
Welcome Back
Sign in to access your dashboard
Username or email
Enter your username or email
Password
Enter your password
Remember me
Sign In

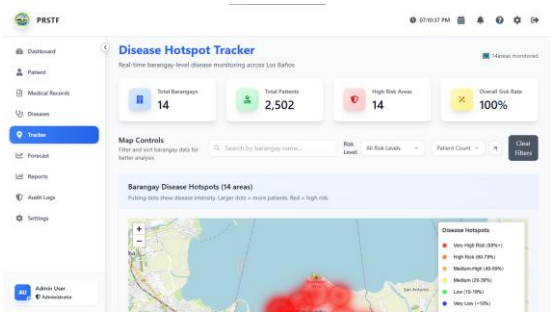
Admin

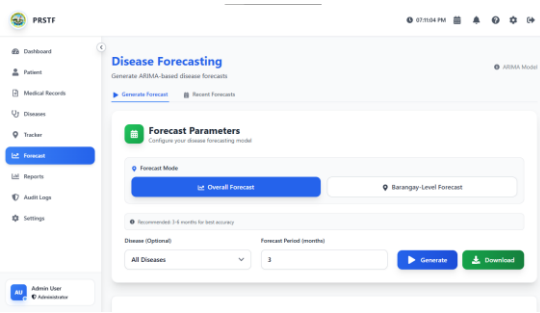

PRSTF
Good Evening, Admin
Dashboard
Total Patients
1,000
Diseases Tracked
5
Active Cases
1,000
New This Month
0
Trends Analysis
Weekly
Monthly
Quarterly
Yearly
Age Distribution
Admin User
Administrator

PRSTF
Patient Records
Manage patient records
Add Patient
Admin User
Administrator

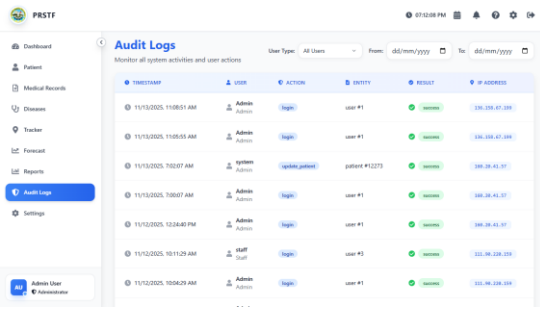

PRSTF
Medical Records
View and manage patient medical records
Admin User
Administrator

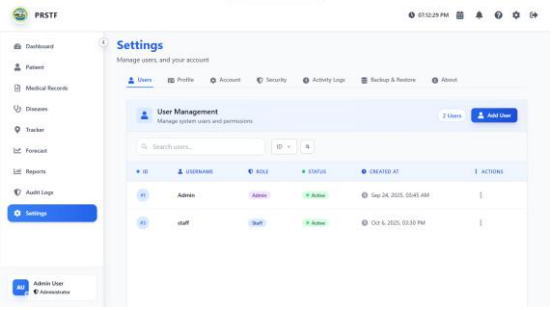

PRSTF
Disease Management
Manage communicable diseases and track cases
Add Disease
Total Diseases
5
Total Cases
2,507
High Risk
0
Case Trends Analysis
Admin User
Administrator

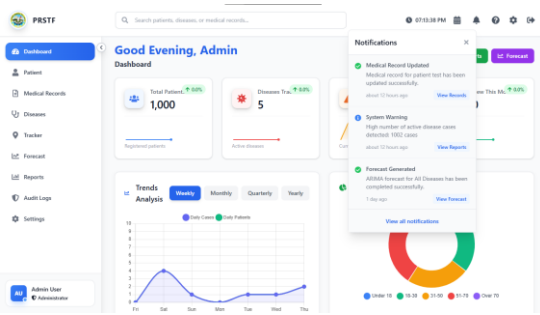

PRSTF
Disease Hotspot Tracker
Real-time barangay-level disease monitoring across Los Baños
Total Barangays
14
Total Patients
2,502
High Risk Areas
14
Overall Sick Rate
100%
Map Controls
Barangay Disease Hotspots (14 areas)
Admin User
Administrator

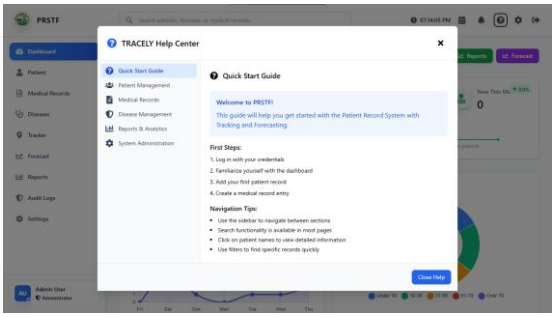

PRSTF
Disease Forecasting
Generate ARIMA-based disease forecasts
Forecast Parameters
Configure your disease forecasting model
Forecast Mode
Overall Forecast
Barangay-Level Forecast
Disease (Optional)
All Diseases
Forecast Period (months)
Generate
Download


PRSTF
Disease Reports & Analytics
Comprehensive health data analysis and insights
Filters
Export CSV
Total Diseases
5
Total Cases
40
High Risk Areas
14
Avg Cases/Day
1.3
Disease Distribution
Gender Distribution

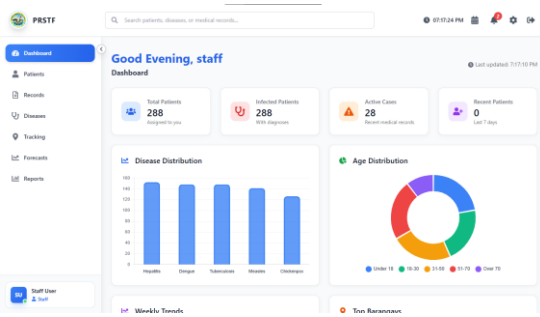

PRSTF
Audit Logs
Monitor all system activities and user actions
TIMESTAMP
USER
ACTION
ENTITY
RESULT
IP ADDRESS

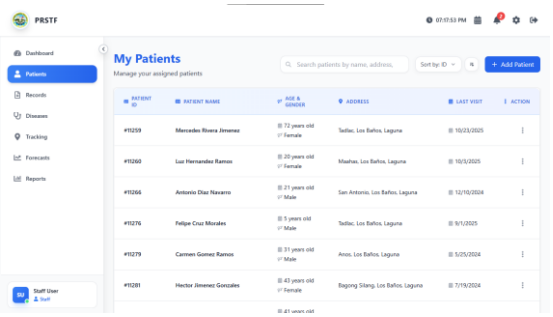

PRSTF
Settings
Manage users, and your account
Users
Profile
Account
Security
Activity Logs
Backup & Restore
About
User Management
Manage system users and permissions
Add User

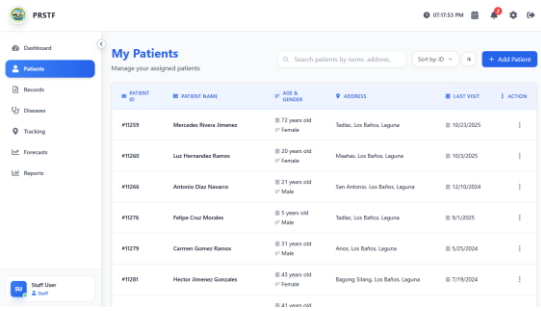

PRSTF
Good Evening, Admin
Dashboard
Total Patients
1,000
5
Trends Analysis
Weekly
Monthly
Quarterly
Yearly
Notifications
Medical Record Updated
System Warning
Forecast Generated
View all notifications

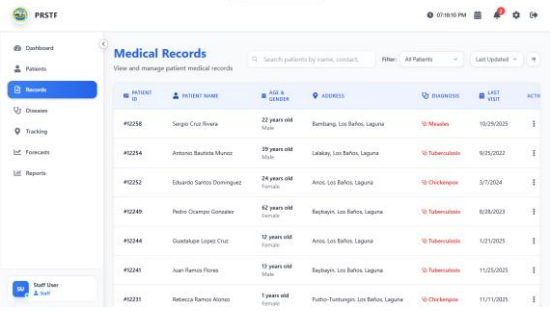

PRSTF
TRACELY Help Center
Quick Start Guide
Patient Management
Medical Records
Disease Management
Reports & Analytics
System Administration
Quick Start Guide
Welcome to PRSTF!
This guide will help you get started with the Patient Record System with Tracking and Forecasting.
First Steps:
1. Log in with your credentials
2. Familiarize yourself with the dashboard
3. Add your first patient record
4. Create a medical record entry
Navigation Tips:
Use the sidebar to navigate between sections
Search functionality is available in most pages
Click on patient names to view detailed information
Use filters to find specific records quickly
Close Help

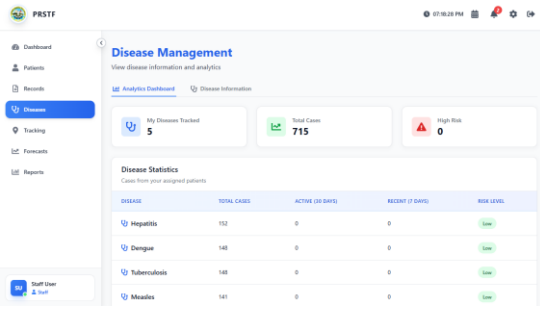
Staff

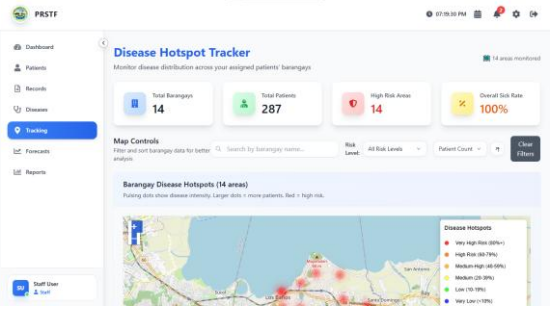

PRSTF
Good Evening, staff
Dashboard
Total Patients
288
Infected Patients
288
Active Cases
28
Recent Patients
0
Disease Distribution
Age Distribution

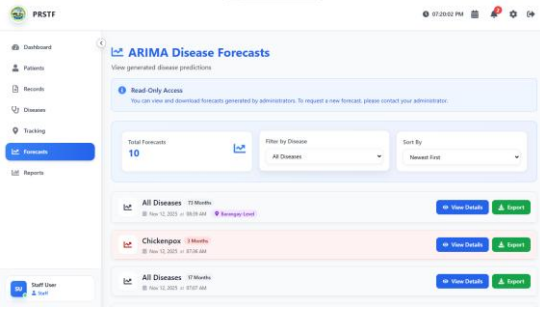

PRSTF
My Patients
Manage your assigned patients
Add Patient

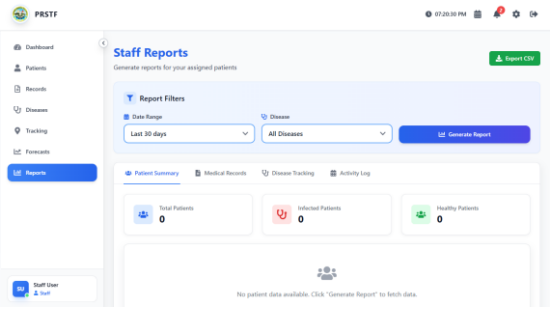

PRSTF
My Patients
Manage your assigned patients
Add Patient

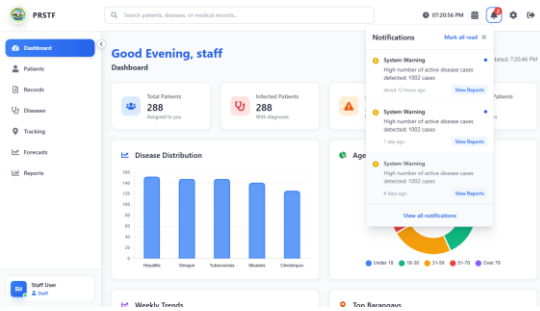

PRSTF
Medical Records
View and manage patient medical records

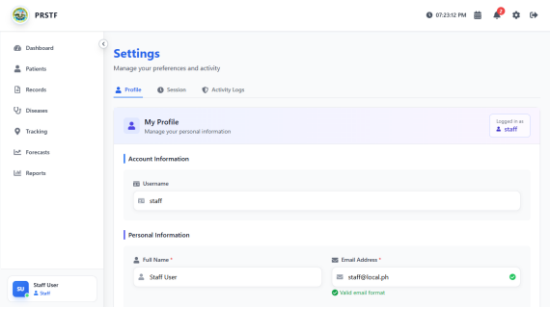

PRSTF
Disease Management
View disease information and analytics
My Diseases Tracked
5
Total Cases
715
High Risk
0
Disease Statistics


PRSTF
Disease Hotspot Tracker
Monitor disease distribution across your assigned patients' barangays
Total Barangays
14
Total Patients
287
High Risk Areas
14
Overall Sick Rate
100%
Map Controls
Barangay Disease Hotspots (14 areas)


PRSTF
ARIMA Disease Forecasts
View generated disease predictions
Read-Only Access
Total Forecasts
10
All Diseases
Chickenpox
All Diseases


PRSTF
Staff Reports
Generate reports for your assigned patients
Export CSV
Report Filters
Last 30 days
All Diseases
Generate Report


PRSTF
Good Evening, staff
Dashboard
Notifications
Mark all read
System Warning
High number of active disease cases detected: 1002 cases
View all notifications
Disease Distribution


PRSTF
Settings
Manage your preferences and activity
Profile
Session
Activity Logs
My Profile
Account Information
Personal Information

APPENDIX F.

Curriculum Vitae



Name: Ariel Longa

Birthdate: January 6, 2001

Address: Los Baños, Laguna

Birthplace: Ligao, Albay

Civil Status: Single

Religion: Roman Catholic

Education:

Elementary: Don Teotimo Elementary School

High School: Ligao City National Technical Vocational

Senior High School: Ligao City National Technical Vocational

Tertiary: Laguna State Polytechnic University Los Baños

Course: Bachelors of Science in Information Technology

Specialization: Major in Web and Mobile Application Development



Name: Mark Brian Navarro

Birthdate: May 26, 2004

Address: Los Baños, Laguna

Birthplace: Oas, Albay

Civil Status: Single

Religion: Roman Catholic

Education:

Elementary: Mabayawas Elementary School

High School: St. Michael's Academy of Oas Inc.

Senior High School: Colegio de Los Baños

Tertiary: Laguna State Polytechnic University Los Baños

Course: Bachelors of Science in Information Technology

Specialization: Major in Web and Mobile Application Development



Name: Ryan Erick Faustino

Birthdate: February 06, 2004

Address: 478 Masiit, Calauan, Laguna

Birthplace: San Pablo City, Laguna

Civil Status: Single

Religion: Roman Catholic

Education:

Elementary: Masiit Elementary School

High School: Liceo de Bay

Senior High School: Liceo de Bay

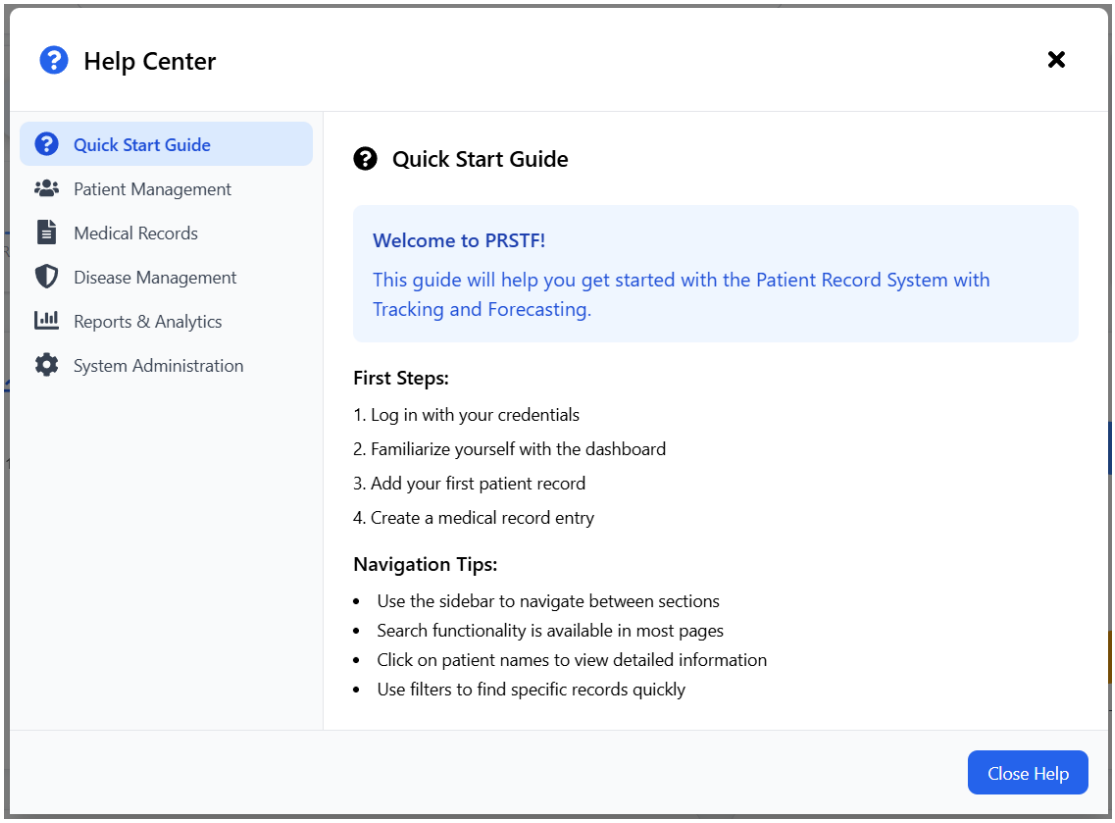
Tertiary: Laguna State Polytechnic University Los Baños

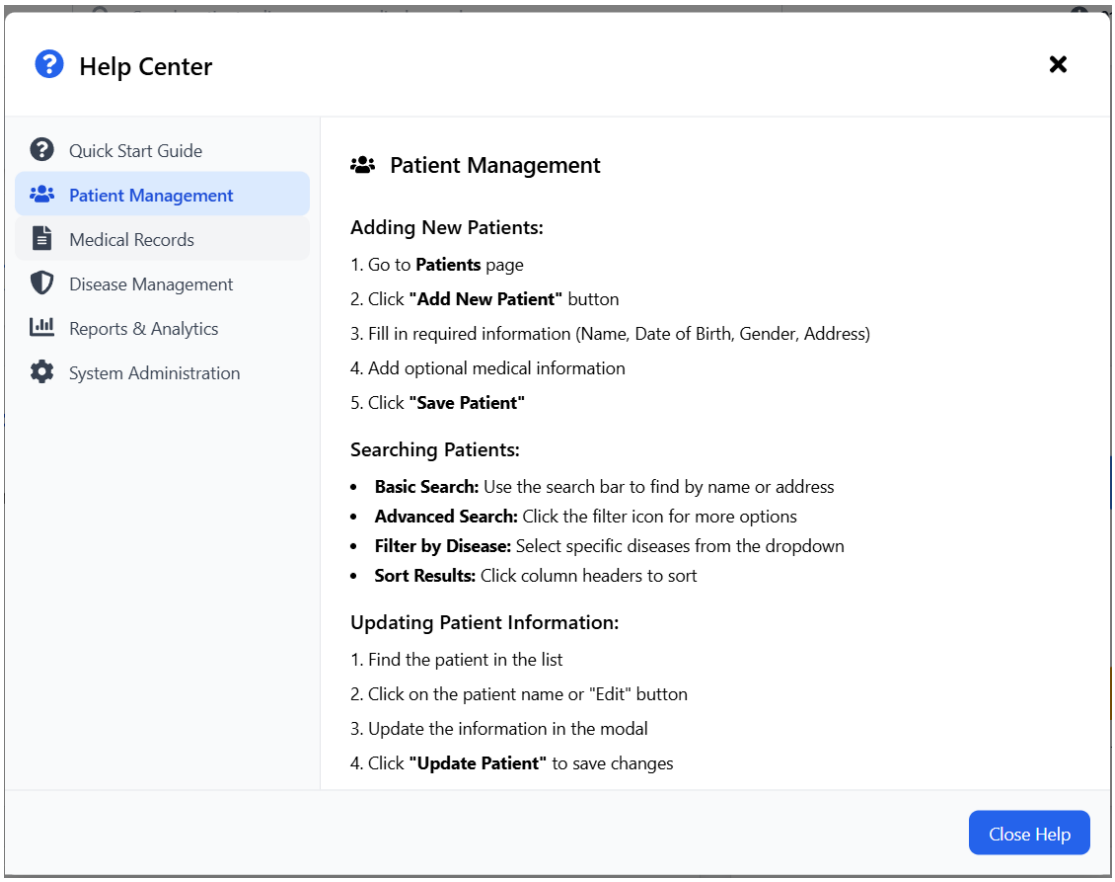
Course: Bachelor of Science in Information Technology

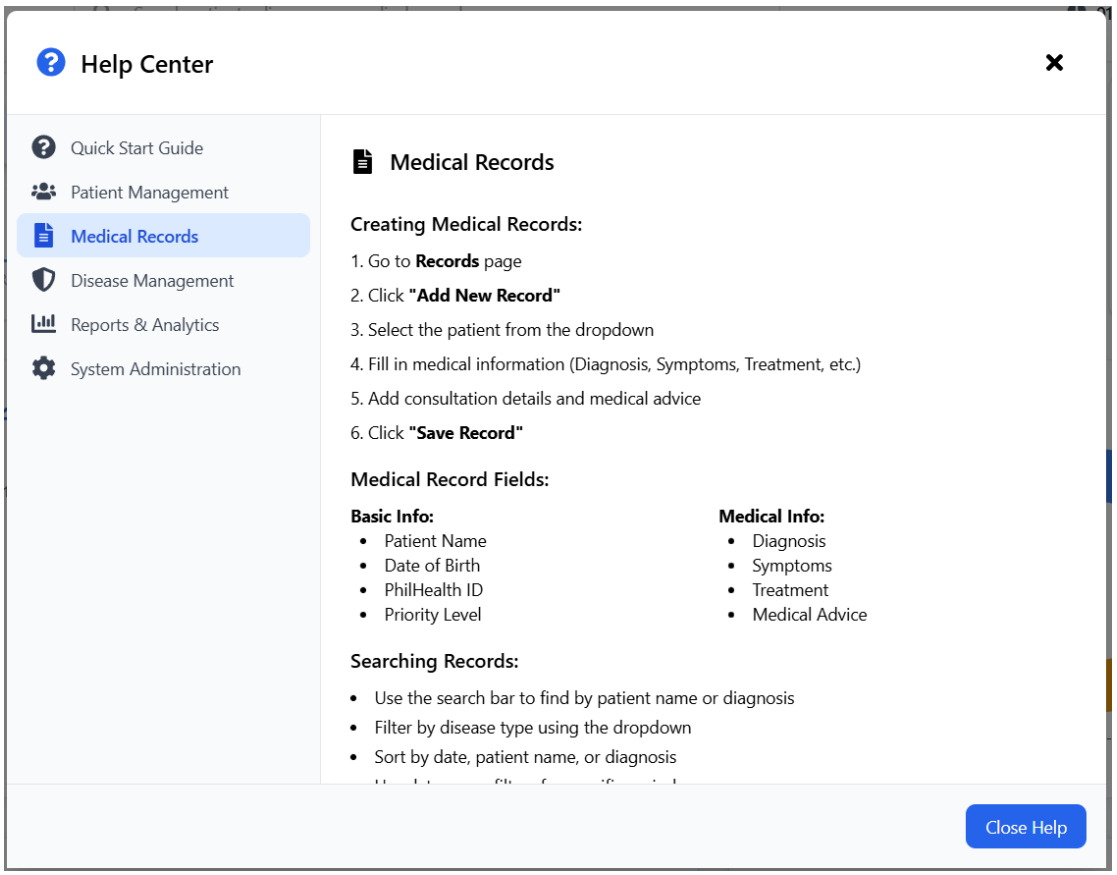
Specialization: Web and Mobile Application Development

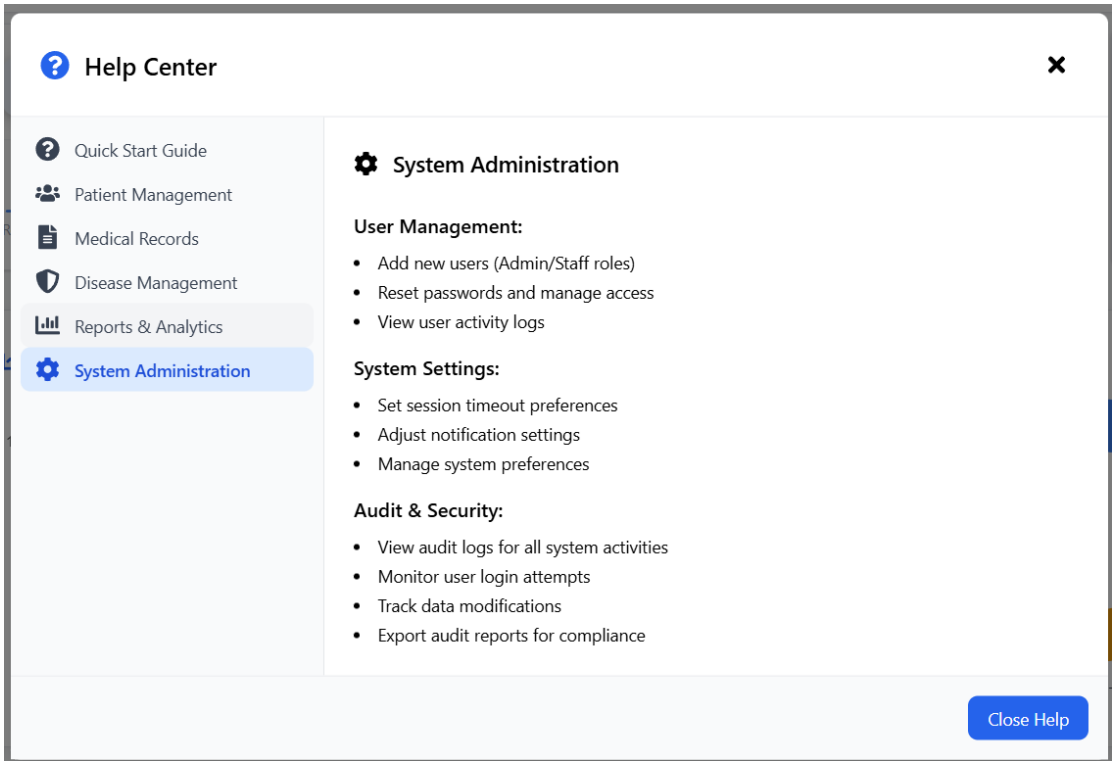
APPENDIX G.

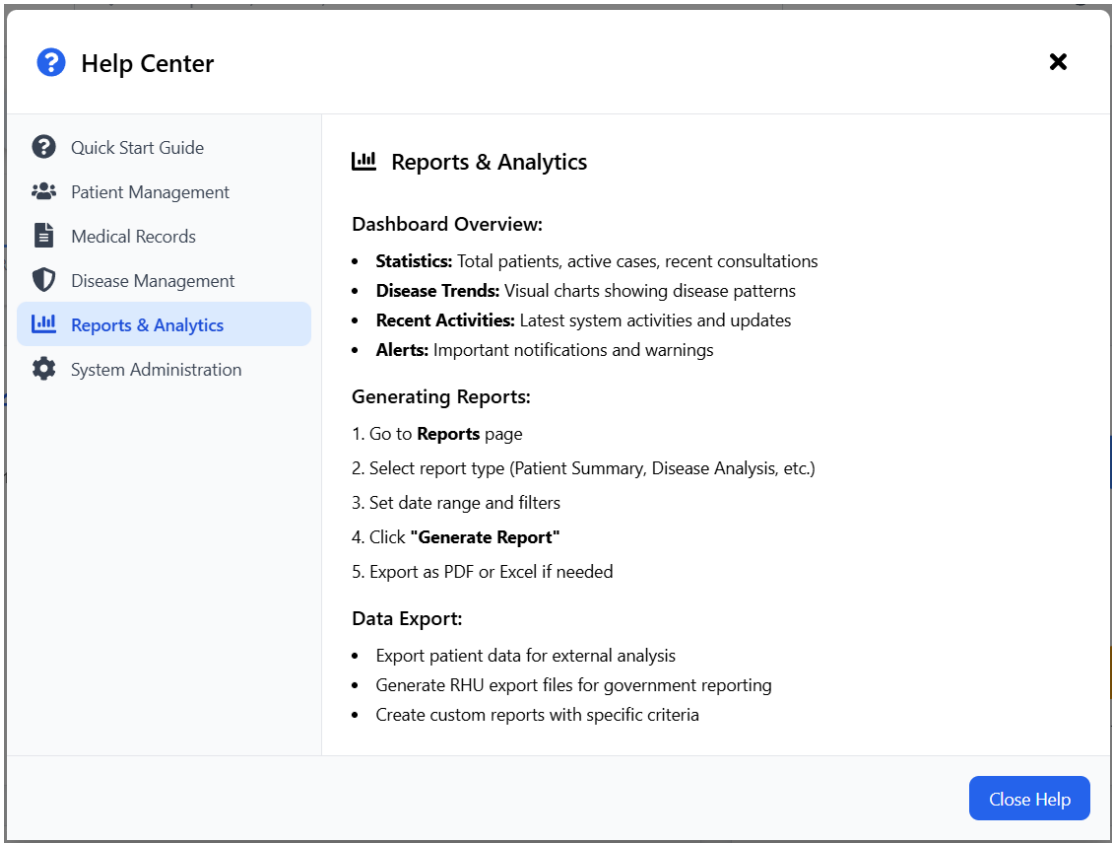
User's Guide

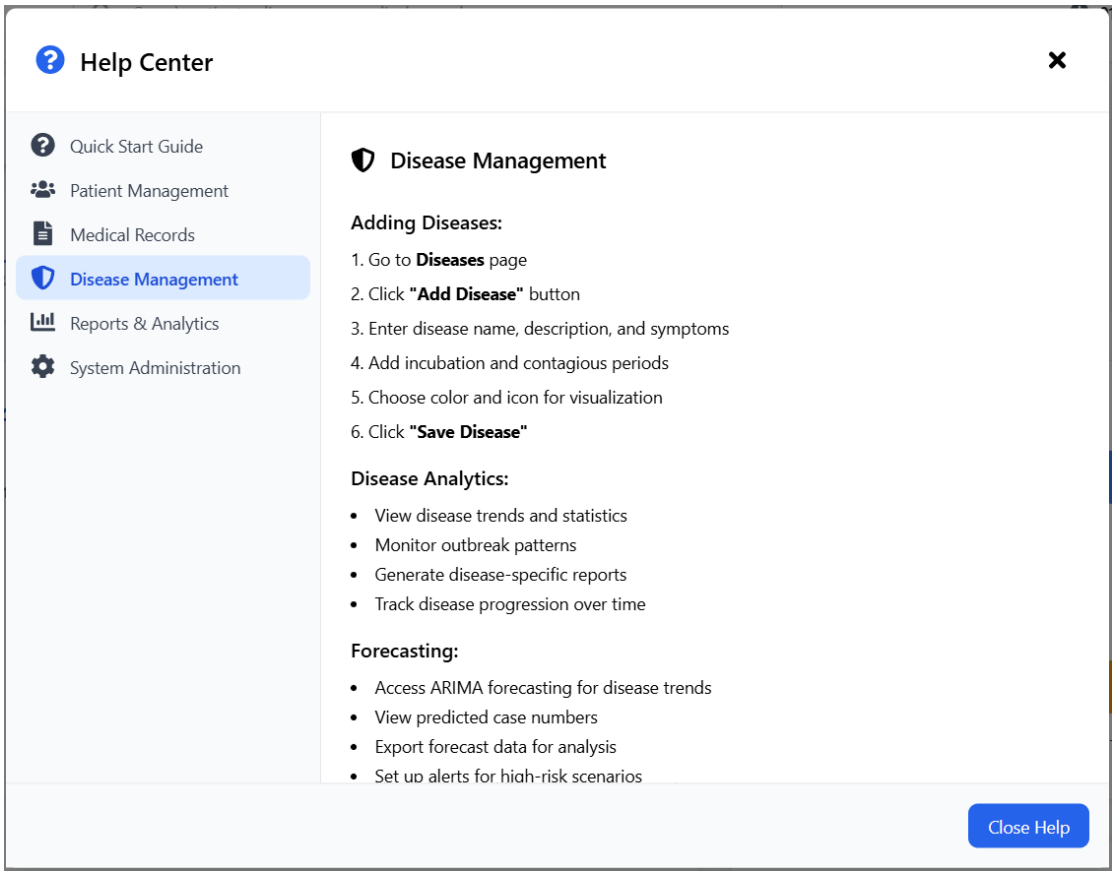

Help Center
Quick Start Guide
Patient Management
Medical Records
Disease Management
Reports & Analytics
System Administration
Quick Start Guide
Welcome to PRSTF!
This guide will help you get started with the Patient Record System with Tracking and Forecasting.
First Steps:
1. Log in with your credentials
2. Familiarize yourself with the dashboard
3. Add your first patient record
4. Create a medical record entry
Navigation Tips:
Use the sidebar to navigate between sections
Search functionality is available in most pages
Click on patient names to view detailed information
Use filters to find specific records quickly
Close Help


Help Center
Quick Start Guide
Patient Management
Medical Records
Disease Management
Reports & Analytics
System Administration
Patient Management
Adding New Patients:
1. Go to Patients page
2. Click "Add New Patient" button
3. Fill in required information (Name, Date of Birth, Gender, Address)
4. Add optional medical information
5. Click "Save Patient"
Searching Patients:
Basic Search: Use the search bar to find by name or address
Advanced Search: Click the filter icon for more options
Filter by Disease: Select specific diseases from the dropdown
Sort Results: Click column headers to sort
Updating Patient Information:
1. Find the patient in the list
2. Click on the patient name or "Edit" button
3. Update the information in the modal
4. Click "Update Patient" to save changes
Close Help


Help Center
Quick Start Guide
Patient Management
Medical Records
Disease Management
Reports & Analytics
System Administration
Medical Records
Creating Medical Records:
1. Go to Records page
2. Click "Add New Record"
3. Select the patient from the dropdown
4. Fill in medical information (Diagnosis, Symptoms, Treatment, etc.)
5. Add consultation details and medical advice
6. Click "Save Record"
Medical Record Fields:
Basic Info:
Patient Name
Date of Birth
PhilHealth ID
Priority Level
Medical Info:
Diagnosis
Symptoms
Treatment
Medical Advice
Searching Records:
Use the search bar to find by patient name or diagnosis
Filter by disease type using the dropdown
Sort by date, patient name, or diagnosis
Close Help


Help Center
Quick Start Guide
Patient Management
Medical Records
Disease Management
Reports & Analytics
System Administration
System Administration
User Management:
Add new users (Admin/Staff roles)
Reset passwords and manage access
View user activity logs
System Settings:
Set session timeout preferences
Adjust notification settings
Manage system preferences
Audit & Security:
View audit logs for all system activities
Monitor user login attempts
Track data modifications
Export audit reports for compliance
Close Help


Help Center
Quick Start Guide
Patient Management
Medical Records
Disease Management
Reports & Analytics
System Administration
Reports & Analytics
Dashboard Overview:
Statistics: Total patients, active cases, recent consultations
Disease Trends: Visual charts showing disease patterns
Recent Activities: Latest system activities and updates
Alerts: Important notifications and warnings
Generating Reports:
1. Go to Reports page
2. Select report type (Patient Summary, Disease Analysis, etc.)
3. Set date range and filters
4. Click "Generate Report"
5. Export as PDF or Excel if needed
Data Export:
Export patient data for external analysis
Generate RHU export files for government reporting
Create custom reports with specific criteria
Close Help


Help Center
Quick Start Guide
Patient Management
Medical Records
Disease Management
Reports & Analytics
System Administration
Disease Management
Adding Diseases:
1. Go to Diseases page
2. Click "Add Disease" button
3. Enter disease name, description, and symptoms
4. Add incubation and contagious periods
5. Choose color and icon for visualization
6. Click "Save Disease"
Disease Analytics:
View disease trends and statistics
Monitor outbreak patterns
Generate disease-specific reports
Track disease progression over time
Forecasting:
Access ARIMA forecasting for disease trends
View predicted case numbers
Export forecast data for analysis
Set up alerts for high-risk scenarios
Close Help

APPENDIX H.

Publishable Paper

PATIENT RECORD SYSTEM WITH DISEASE TRACKING AND ARIMA-BASED FORECASTING FOR TOP 5 DISEASES IN LOS BAÑOS, LAGUNA

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Abstract* — *Rural Health Units in the Philippines continue to face challenges in managing patient records and monitoring disease trends due to manual and paper based processes, resulting in delayed reporting data loss and limited support for timely decision making. This study presents the development of a Patient Record System with Disease Tracking and ARIMA Based Forecasting designed to improve record management disease monitoring and short term prediction for the top five diseases in Los Baños Laguna. The system provides a web based platform that enables health workers to manage patient information, visualize disease distribution using GIS mapping and forecast future case trends through an ARIMA model. Developed using the Agile Scrum methodology and evaluated using ISO 25010 and the Technology Acceptance Model, the system achieved good ratings in functionality, usability, reliability and user acceptance, indicating its suitability for real Rural Health Unit use and improved disease surveillance and healthcare delivery.

Keywords: *Patient Record System, Disease Tracking, ARIMA Forecasting, GIS, Rural Health Unit*

1. INTRODUCTION

Diseases remain a major public health concern in developing countries such as the Philippines where Rural Health Units often rely on manual systems for patient record keeping and disease reporting. These traditional methods are prone to errors, delayed reporting and data inconsistency which affect the ability of health workers to monitor disease trends and respond promptly to possible outbreaks. The lack of digital tools also limits the effective use of health data for planning and decision making at the local level.

Recent advancements in health information systems have shown that digital record management combined with geographic and predictive analytics can significantly improve disease surveillance and healthcare operations. Geographic Information Systems allow visualization of disease cases across locations while time series forecasting models such as ARIMA enable the prediction of short-term disease trends based on historical data. Integrating these technologies into a single system can provide health workers with timely and actionable insights.

This study aims to address these challenges by developing a Patient Record System with Disease Tracking and ARIMA Based Forecasting for the top five diseases recorded in Los Baños Laguna. The system is intended to support Rural Health Units by improving patient data management, enhancing disease monitoring and providing short term forecasts to aid in health planning and response.

2. METHODOLOGY

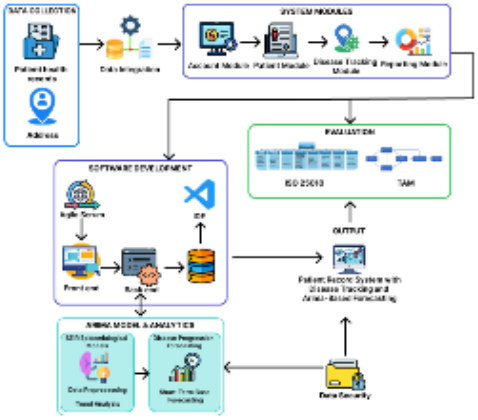
Figure 1 presents the overall process used in the development of the Patient Record System with Disease Tracking and ARIMA Based Forecasting. The study was guided by related local and foreign literature that provided background information and technical direction for system development. Existing works on

health information systems disease surveillance geographic information systems and time series forecasting were reviewed to support the design and implementation of the system. These references helped identify common challenges in Rural Health Units particularly in managing patient records monitoring disease cases and predicting possible trends. The integration of digital record management GIS visualization and forecasting techniques aims to improve reporting efficiency planning and decision making in local health facilities.

Figure 1. Conceptual Framework

2.1 PROJECT DESIGN

The system was designed using standard architectural practices to meet the operational requirements of Rural Health Units. The architecture shown in Figure 2 is composed



of interconnected modules that work together to support patient management disease monitoring and forecasting. The Account Module handles user authentication access control profile management and logout to ensure secure system usage. The Patient Module allows health workers to register patients, update records and view medical history to maintain organized and accurate health data. The Disease Tracking Module supports recording of disease cases, monitoring case status and mapping case locations to visualize disease spread within barangays. The Reporting Module enables users to generate reports, view summarized dashboards, retrieve alerts and export data for documentation and analysis. The ARIMA Based Analytics Module processes historical disease data to identify trends and produce short term forecasts that assist health administrators in monitoring and responding to possible outbreaks.

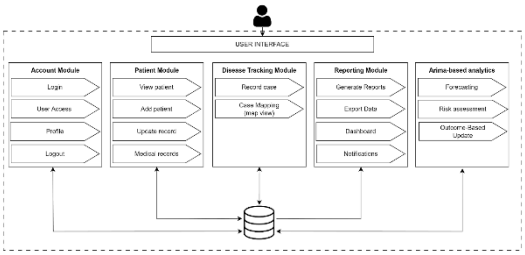
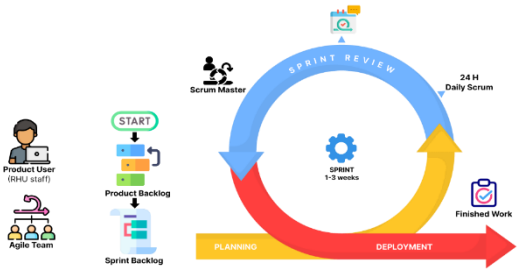


Figure 2. System Architecture

2.2 PROJECT DEVELOPMENT

Figure 3 shows the Agile Scrum methodology applied during system development. The Agile approach was selected to allow continuous feedback and gradual improvement of the system. Development was divided into multiple sprint cycles lasting one to four weeks where each sprint focused on delivering specific system features. Activities such as product backlog creation sprint planning, development testing and review were performed in each cycle to ensure that system requirements were addressed and refined based



on user feedback.

Figure 3. Agile Scrum

2.3 SOFTWARE TOOLS

MySQL serves as the main database of the system and is used to store patient information, disease case records and forecasting outputs.

PHP is used to handle backend operations including database communication and system logic.

Node.js supports asynchronous processing and API communication between system components.

Vite React.js is used to develop the user interface allowing the creation of responsive

and interactive system pages such as dashboards forms and tracking views.

Python is utilized to implement the ARIMA based forecasting model using historical disease case data.

Tailwind CSS is applied to manage the layout design and responsiveness of the interface.

HTML provides the basic structure of all system pages.

2.4 EVALUATION PROCEDURE

The quality of the developed system was evaluated using the ISO 25010 software quality standard. This standard was used to assess key quality attributes including functionality, usability, reliability and efficiency. IT experts conducted beta testing to evaluate the system based on both functional and non functional aspects such as system structure design and performance. A five point Likert scale ranging from 5 – Excellent, 4 – Good, 3 – Fair, 2 – Needs Improvement, and 1 – Poor was used to rate each system module and feature. And continuous testing during development helped identify issues early and ensured that the system met quality requirements before deployment.



Figure 4. Quality Characteristic of ISO 25010

Technology Acceptance Model (TAM)

The Technology Acceptance Model was applied to evaluate user acceptance of the system. This model was used to measure users' perceptions in terms of ease of use, usefulness, attitude toward using and intention to use the system. Survey questionnaires were distributed to Rural Health Unit staff and administrators who regularly interact with the system. The results provided insight into how well the system supports daily tasks and overall user satisfaction.

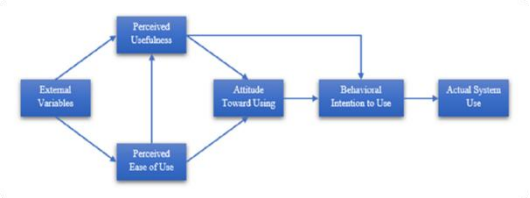


Figure 5. Technology Acceptance Model

Project Testing and Evaluation

Project testing and evaluation were conducted to ensure that the Patient Record System with Disease Tracking and ARIMA-Based Forecasting operates correctly and meets the defined system requirements. Testing focused on verifying that each module functions properly both independently and when integrated with other components. The developers performed systematic testing during development to identify possible errors, bugs, and performance issues early in the process.
Software testing was carried out to confirm that system features such as patient record management, disease tracking, reporting, GIS visualization, and forecasting operate as intended. Unit testing was performed to validate individual modules and ensure that each function produces correct and expected results. Integration testing followed to verify that data flows smoothly between modules and that the system performs consistently based on the defined workflow. Feedback gathered during testing was used to refine system performance and usability before final deployment.

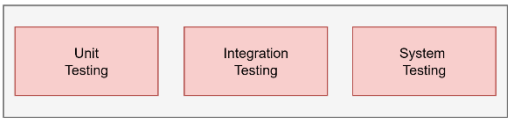


Figure 6. Stages of Testing

Dataset, Model Development, and Evaluation

The dataset used in this study was obtained from the Rural Health Unit and consists of aggregated disease case records from 2021 to 2024. The data includes barangay, age group, sex, disease type, and total reported cases, focusing on measles, chickenpox, hepatitis, tuberculosis, and dengue. Patient-level identifiers were excluded in compliance with the Data Privacy Act of 2012. To support system testing, synthetic patient data with

fictional details was generated based on observed patterns, while only real aggregated data was used for ARIMA forecasting.
Data preprocessing was conducted to ensure accuracy and consistency. Records were reviewed for missing or inconsistent values and organized into a monthly time-series format. The dataset was divided into training and testing sets using an 80 percent and 20 percent split to support model development and validation.

The forecasting model was developed using the ARIMA approach based on historical disease case trends. Model parameters were selected through analysis of training data patterns, and multiple configurations were tested to identify the best-performing model. Model evaluation was performed using Mean Absolute Error and Root Mean Square Error by comparing forecasted values with actual case records. These results confirmed that the ARIMA model can support short-term disease trend analysis and planning.

Population of the Study

The respondents of the study consisted of Information Technology experts and Rural Health Unit personnel involved in the evaluation and use of the Patient Record System with Disease Tracking and ARIMA-Based Forecasting. These respondents represent both technical evaluators and actual system users, ensuring that the assessment covered system quality, usability, and acceptance in a real Rural Health Unit setting.

Sampling Design

The study utilized purposive sampling to select respondents who were directly involved in patient record management, disease monitoring, and system operation at the Rural Health Unit. Three (3) Information Technology experts were chosen to evaluate the system based on the ISO 25010 software quality model, focusing on functionality, usability, reliability, and efficiency. For user acceptance evaluation using the Technology Acceptance Model, a total of one hundred (100) respondents participated, consisting of seven (7) administrators who manage the system and ninety-three (93) Rural Health Unit and local health staff members who actively use it. This sampling approach ensured that feedback was obtained from individuals with relevant knowledge and hands-on experience with the developed system.

Data Collection Instrument

Data were collected through interviews and structured survey questionnaires administered to the respondents. The gathered data served as the basis for evaluating the performance and effectiveness of the Patient Record System with Disease Tracking and ARIMA-Based Forecasting. The primary data gathering tool was a structured survey questionnaire designed based on the ISO 25010 software quality model and the Technology Acceptance Model. The questionnaire was divided into sections covering the demographic profile of respondents and system evaluation criteria, including functionality, usability, reliability, efficiency, compatibility, security, maintainability, and portability. Responses were measured using a five-point Likert scale ranging from 1 (Poor) to 5 (Excellent). Open-ended questions were also included to obtain additional feedback on system usability and possible improvements.

3. RESULTS AND DISCUSSION

This section presents the results of the evaluation of the Patient Record System with Disease Tracking and ARIMA-Based Forecasting based on the objectives of the study. The system was assessed in terms of patient record management, disease tracking, forecasting capability, software quality, and user acceptance in a Rural Health Unit setting.

Patient and Medical Record Management

The developed system includes a secure authentication mechanism that requires authorized Rural Health Unit personnel to log in before accessing any system feature. User roles are defined to control access levels for administrators and staff, ensuring data confidentiality and preventing unauthorized modification of patient information. The Patient

Module consolidates all registered patients into a searchable and sortable table, allowing staff to efficiently view and retrieve patient records. A guided data entry form is used when adding new patients, clearly indicating required and optional fields to ensure complete and consistent record encoding.

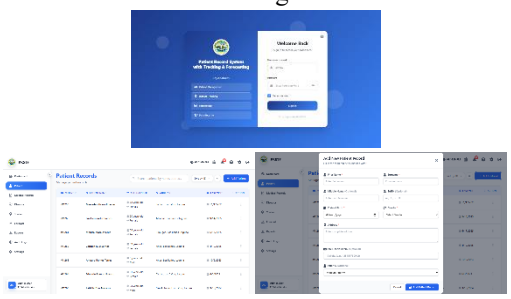


Figure 7. Log In and Patient Module

The Medical Records module provides structured access to diagnosis summaries, visit history, and clinical notes. Selecting a patient opens a detailed medical profile containing demographic data, consultation history, and vital information, supporting continuity of care and accurate clinical decision making. These features provide a secure and organized approach to patient record storage, retrieval, and updating.

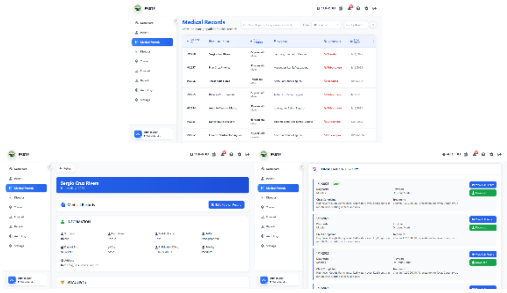


Figure 7. Medical Records Module

Map-Based Disease Tracking

The system implements a Disease Tracking module that visualizes the distribution of communicable disease cases across barangays in Los Baños. The interface displays summary indicators such as total patients, monitored barangays, and identified high-risk areas, providing a quick overview of the current disease situation. Filtering tools allow users to view cases by disease type, barangay, or risk level. A GIS-based map presents color-coded markers representing case density, highlighting potential hotspots. This visual representation supports RHU personnel in identifying patterns, monitoring outbreaks, and planning targeted public health interventions.

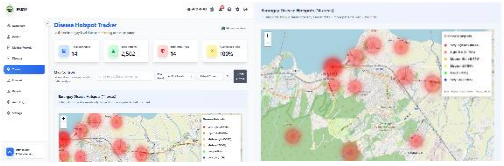


Figure 7. Disease Tracking Module

ARIMA-Based Disease Forecasting

An ARIMA-based forecasting module was developed to generate short-term predictions for the top five recorded diseases. Users may configure forecast parameters including disease selection, forecast duration, and scope at the overall or barangay level.

The system produces a forecast summary showing recent case counts, projected trends, and interpretative insights to support planning and resource allocation. Detailed outputs include monthly predicted values and accuracy metrics such as RMSE and MAE. Graphical visualizations comparing actual and forecasted data assist users in interpreting trends and evaluating prediction reliability. Barangay-level forecasting further supports localized preparedness and monitoring.

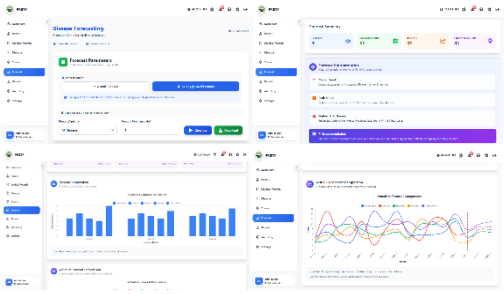


Figure 7. Medical Records Module

Project Testing and Debugging

Project testing and debugging were conducted to ensure that the Patient Record System with Disease Tracking and ARIMA-Based Forecasting functions correctly, integrates smoothly, and performs reliably in actual RHU operations.

Unit testing confirmed that all core modules of the system function correctly when tested

individually. User authentication patient management medical records disease tracking forecasting reports and user account management performed as expected with all tests passing and no functional errors observed.

Integration testing verified that system modules work properly when combined. Connections between login dashboard patient records, medical records and user roles operated smoothly ensuring accurate data flow consistent updates and correct access control across the system.

System testing validated the overall stability and reliability of the system under real use conditions. All modules operated continuously without errors data remained consistent across features and forecasting and reporting functions executed correctly confirming that the system is ready for deployment in Rural Health Unit operations.

Forecast Accuracy Metrics of the ARIMA Model

The forecasting performance of the ARIMA model was assessed using Mean Absolute Error and Root Mean Square Error. Chickenpox produced low error values with an RMSE of 2.52 and MAE of 2.10 indicating reliable short term predictions. Dengue recorded an RMSE of 2.62 and MAE of 2.00 showing minimal difference between predicted and actual case counts. Hepatitis obtained an RMSE of 2.81 and MAE of 2.60 reflecting acceptable error levels and close alignment with observed trends. Measles showed higher error values with an RMSE of 4.88 and MAE of 4.70 due to its more irregular case pattern but the model was still able to capture the general movement of cases. Tuberculosis achieved an RMSE of 2.59 and MAE of 2.20 demonstrating good forecast accuracy with predictions close to actual values. These results show that the ARIMA model performs well for most diseases and supports short term disease trend analysis in Rural Health Unit operations.

ISO 25010 Evaluation

The system was evaluated using the ISO 25010 software quality model by three Information Technology experts through a 5 point Likert scale. The results show that the system achieved a composite mean of 4.89 interpreted as Excellent indicating high overall software quality. Functional suitability, security and flexibility obtained the highest mean ratings of 5.00 while performance efficiency, usability and reliability recorded mean values of 4.93 to 4.94. Compatibility, maintainability and safety also received excellent ratings with mean values of 4.67 4.73 and 4.80 respectively. The low standard deviation values indicate consistent expert assessment across all characteristics. These results confirm that the system meets ISO 25010 quality standards and is suitable for deployment and continued use in Rural Health Unit operations.

Table 1. ISO 25010 Evaluation Results

Characteristics	Mean	Standard Deviation	Interpretation
Functional Suitability	5.00	0.00	Excellent
Performance Efficiency	4.93	0.12	Excellent
Compatibility	4.67	0.47	Excellent
Usability	4.94	0.12	Excellent
Reliability	4.93	0.12	Excellent
Security	5.00	0.00	Excellent
Maintainability	4.73	0.25	Excellent
Flexibility	5.00	0.00	Excellent
Safety	4.80	0.26	Excellent
Composite Mean	**4.89**		**Excellent**

Legend: 1–1.80 = Poor, 1.81–2.60 = Fair, 2.61–3.40 = Good, 3.41–4.20 = Very Good, 4.21–5.00 = Excellent

n=3 IT experts

Technology Acceptance Model (TAM) Results

The acceptability of the system was assessed using the Technology Acceptance Model with a total of 100 respondents composed of administrators and Rural Health Unit staff. The evaluation focused on perceived usefulness, perceived ease of use, attitude toward using, and behavioral intention to use, using a 4-point Likert scale. The results show that perceived usefulness obtained a mean of 3.55, perceived ease of use 3.56, attitude toward using 3.55, and behavioral intention to use 3.57, all interpreted as Highly Acceptable. The composite mean of 3.56 indicates strong user acceptance, showing that users find the system

useful, easy to operate, reliable, and are willing to continue using it in actual health center operations.

Table 2. Summary of TAM Evaluation Results

TAM Construct	Mean	Standard Deviation	Interpretation
Perceived Usefulness	3.55	0.56	Highly Acceptable
Perceived Ease of Use	3.56	0.55	Highly Acceptable
Attitude Toward Using	3.55	0.55	Highly Acceptable
Behavioral Intention to Use	3.57	0.54	Highly Acceptable
Composite Mean	**3.56**		**Highly Acceptable**

Legend: 1.00–1.75 = Not Acceptable, 1.76–2.50 = Slightly Acceptable, 2.51–3.25 = Acceptable, 3.26–4.00 = Highly Acceptable.

4. SUMMARY, CONCLUSION, AND RECOMMENDATION

This study developed a web-based Patient Record System with Disease Tracking and ARIMA-Based Forecasting to support Rural Health Units in improving patient record management disease monitoring and short-term forecasting. The system was created to address challenges in manual record keeping, delayed reporting and limited analytical support commonly experienced in local health centers. Core modules include digital patient records map-based disease tracking for the top five communicable diseases namely measles chickenpox hepatitis tuberculosis and dengue and an ARIMA-based forecasting module for short-term trend prediction. The system supports improved data accuracy accessibility and timely decision making while aligning with SDG 3 Good Health and Well-Being SDG 9 Industry Innovation and Infrastructure and SDG 11 Sustainable Cities and Communities.

Conclusion

The Patient Record System with Disease Tracking and ARIMA-Based Forecasting achieved its intended objectives by providing an organized digital platform for patient data management enhanced disease surveillance and basic forecasting support. The patient module reduced reliance on paper records and improved data retrieval while the disease tracking module enabled clearer visualization of case distribution and hotspots. The ARIMA forecasting feature provided useful short-term insights to support planning and preparedness. System evaluation using ISO 25010 and user feedback indicated that the platform meets key quality requirements and is acceptable for real RHU operations. Overall the system demonstrates the value of digital health solutions in strengthening local disease monitoring and healthcare delivery.

Recommendations

The following recommendations are proposed to support future enhancement and wider application of the Patient Record System with Disease Tracking and ARIMA-Based Forecasting.

Extend system implementation to barangay health centers. The system may be deployed beyond Rural Health Units to barangay health centers to reduce manual record keeping and strengthen community-level disease monitoring.

Implement a centralized database. A centralized data repository can support real-time data sharing, improve reporting efficiency and enable coordinated health responses across facilities.

Enhance forecasting through advanced models. Future development may incorporate advanced forecasting models such as SARIMA or hybrid approaches to improve prediction accuracy and handle seasonal disease patterns.

Expand disease and geographic coverage. Including additional communicable diseases and broader location coverage can provide more comprehensive trend analysis and support targeted health interventions.

Use more detailed datasets. Access to more complete and accurate case data while complying with data privacy regulations can further improve forecasting reliability and decision-making support.

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APPENDIX I.

Plagiarism Results

PATIENT RECORD SYSTEM WITH DISEASE TRACKING AND ARIMA-BASED FORECASTING FOR TOP 5 DISEASES IN LOS BAÑOS, LAGUNA

by Ronah joy Ramirez

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